PALEOMAGNETISM OF CRETACEOUS AND PALEOCENE SEDIMENTARY ROCKS ACROSS THE CASTLE MOUNTAIN FAULT, SOUTH CENTRAL ALASKA

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Abstract. Paleomagnetic and rock magnetic analyses on 217 samples from 13 sites in the Paleocene Chickaloon Formation south of the Castle Mountain Fault and 111 samples from 9 sites in the coeval, but lithologically distinct, Arkose Ridge Formation north of this fault indicate that these rocks contain a pre-folding magnetization carried by fine grained (< 1.0 μm) single domain magnetite. Secondary magnetizations are common, possibly as the result of the presence of authigenic or hydrothermal pyrrhotite. Although characteristic magnetizations were isolated for the Chickaloon and Arkose Ridge rocks, the best results were obtained from demagnetization plane analysis which estimates the location of the paleomagnetic pole for the Chickaloon Formation at 50.5°N, 277.2°E, δ_m = 12.2°, δ_p = 7.7°, and a paleomagnetic pole for the Arkose Ridge Formation at 60.4°N, 138.6°E, δ_m = 11.6°, δ_p = 6.4°. These results suggest that there is no paleomagnetically discernible latitudinal offset across the Castle Mountain Fault since Paleocene time, but that both the Chickaloon and Arkose Ridge rocks, as part of the Peninsular terrane, originated approximately 1600 ±1200 km south of their present position with respect to North America. One possible explanation of these data is that the Peninsular terrane was accreted to North America at mid-latitudes in the Cretaceous and was subsequently translated northward by right-lateral strike-slip faulting parallel to the North American margin. Hence, the Arkose Ridge and Chickaloon results may be indicative of the cumulative right-hand displacement occurring on these faults since Paleocene time. However, a calculation using the pole to the small circle fit of the present-day curvature of the Tintina-Northern Rocky Mountain Trench and Denali fault systems, and the maximum amount of structurally estimated offset across these fault systems, indicates that motion on these faults can account for no more than half of the paleomagnetically observed latitudinal displacement. Thus, unless these faults had significantly different orientations in Paleocene time, or the structural analyses have underestimated the amount of slip on these strike-slip faults, the actual latitudinal displacement predicted for the Arkose Ridge and Chickaloon rocks, based on the 95% confidence limits, may be closer to the lower limit of 400 km.

Paleomagnetic and rock magnetic results from 174 samples from 10 sites in the Matanuska Formation indicate that the magnetic signal is also carried by single domain magnetite. The 4 Matanuska sites north of Castle Mountain Fault yielded a paleomagnetic pole at 50°N, 131° E, K = 9.8, suggesting that these rocks were deposited 2880 ±2000 km south of their present latitudinal position with respect to North America. Based on tectonic arguments similar to the one outlined above, these results suggest that either there was rapid motion of the Peninsular terrane in the Late Cretaceous or that like the Chickaloon and Arkose Ridge rocks, the actual amount of latitudinal displacement of the Matanuska Formation is closer to its lower 95% confidence limit of 800 km. South of Castle Mountain Fault, some Matanuska rocks carry only a viscous present-day field direction and others an Eocene PTRM caused by Eocene-age sills intruding into the the Cretaceous and Tertiary strata in the western part of the Matanuska Valley.

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Introduction

Although it is now widely accepted that much of southern Alaska is composed of allochthonous, and in part, exotic terranes, the details of the timing of their accretion to North America and their subsequent tectonic development are still

debated. Structural arguments suggest that a major period of collisional tectonics dominated the Mesozoic history of the northern Cordillera but by the late Cretaceous the region had developed into a simple "Andean type" continental margin [e.g. Jones et al., 1972, Richter and Jones, 1973; or Csejtey, 1976; Davis et al., 1978; Monger and Price, 1979; Coney, 1978 ; Coney et al. 1980]. In this model, the evolution of the region in the late Cretaceous and early Tertiary, following accretion, was simply related to the internal response of this consuming plate margin to the impinging mosaic of Pacific basin plates.

Paleomagnetic data, which were paramount in the development of the accretion model, are apparently in conflict with late Cretaceous/early Tertiary geological data from southern Alaska. South of the Peninsular terrane, in the Prince William and Chugach terranes, paleomagnetic data from Paleocene volcanic rocks on Kodiak Island [Plumley et al., 1983] and a layered gabbro in the St. Elias Range [Gromme and Hillhouse, 1981] suggest that these early Tertiary rocks originated at least 16° south of their present positions. Even within the southern part of the Peninsular terrane on the Alaska Peninsula, paleomagnetic data indicate that this area was as much as 30° south of its present position in the early Tertiary [Stone et al., 1982; Stone and Packer, 1977, 1979]. In contrast, early Tertiary paleomagnetic poles from volcanic rocks within the Peninsular terrane around Iliamna Lake [Thrupp and Coe, 1986], in the Talkeetna Mountains [Hillhouse et al., 1985] in the Cantwell Basin [Hillhouse and Gromme, 1982] and in the McGrath region [Plumley, 1984] suggest that these rocks were all erupted near their current latitudinal position with respect to North America 50 Ma. Moreover, in the latest Cretaceous-earliest Tertiary geologic record of the area between the two regions, there are no obvious tectonic features suggestive of collision; eg. regional dynamothermal metamorphism or ophiolite emplacement.

A striking structural feature in the Tertiary geology of the northern Cordillera is a system of high angle faults that trend subparallel to the continental margin [King, 1969]. On a regional scale, the traces of these faults are concave seaward, some falling on small circle arcs [St. Amand, 1957], and all have either documented or suspected right-lateral strike-slip offsets. Thus, an alternative to an early Tertiary collision along some as yet unrecognized suture zone is a model in which the collision of these southern Alaskan terranes with North America occurred much farther to the south in the middle or late Cretaceous and that tectonic slices within them were then translated northward by regional right-lateral strike-slip faulting in the early Tertiary [Moore et al., 1983; Coe et al., 1985]. This scenario is attractive in that it does not require post-Cretaceous suturing and it allows for the final positioning of the various tectonic domains to be distributed throughout late Mesozoic and early Cenozoic time.

Closer examination of the available paleomagnetic data (Figure 1) reveals that the Castle Mountain-Bruin Bay Fault system divides southern Alaska into two major magnetotectonic regions. Except for the data from MacColl Ridge in the Wrangell Mountains [Panuska, 1985], the paleomagnetic data north of this system show no paleomagnetically detectable transport in post-Paleocene time while all the rocks outboard from these faults appear to have been translated a minimum of 16° northward between the latest Cretaceous and the early Tertiary. Given these observations, it appears that the Castle Mountain/Lake Clark/Bruin Bay Fault system may have been a strike-slip system with major offsets during the Tertiary.

The Castle Mountain fault system extends over 200 km from the Susitna lowlands through the Matanuska Valley to the Copper River Basin and along with the Bruin Bay Fault, the Castle Mountain Fault delimits the northern boundary of the Cook Inlet Basin (Figure 2). This fault system is the physiographic and structural boundary between the Talkeetna Mountains to the north and the middle to western reaches of the Matanuska Valley to the south, juxtaposing two coeval, but lithologically distinct sedimentary units, the Chickaloon and Arkose Ridge formations [Silberman and Grantz, 1984]. In order to test the hypothesis of large-scale horizontal displacement, as evidenced by latitudinal offset across the Castle Mountain Fault, a detailed paleomagnetic study of these Paleocene units as well as the underlying Cretaceous Matanuska formation was performed. These investigations included a series of redeposition and recompaction experiments on disaggregated slurries of the Arkose Ridge rocks to test for possible inclination or compaction shallowing of its depositional remanent magnetization (DRM). Results from this study not only limit the relative latitudinal displacement across the Castle Mountain Fault system in the latest Cretaceous and early Tertiary but place new constraints on the paleomagnetic pole path for this portion of the Alaskan crust.

Geology

The Cretaceous and Tertiary strata of the Matanuska Valley and lower Talkeetna Mountains are part of the thick sequence of sedimentary and volcanic rocks deposited in the Cook Inlet Basin, a large forearc region located just seaward of the Alaskan-Aleutian volcanic arc massif [Kirchner and Lyon, 1973]. The margins of this basin are defined by the Castle Mountain-Bruin Bay fault system to the north and the Border Ranges fault system to the south which converge towards the northeast in the vicinity of the Matanuska Valley. This valley forms the narrowed

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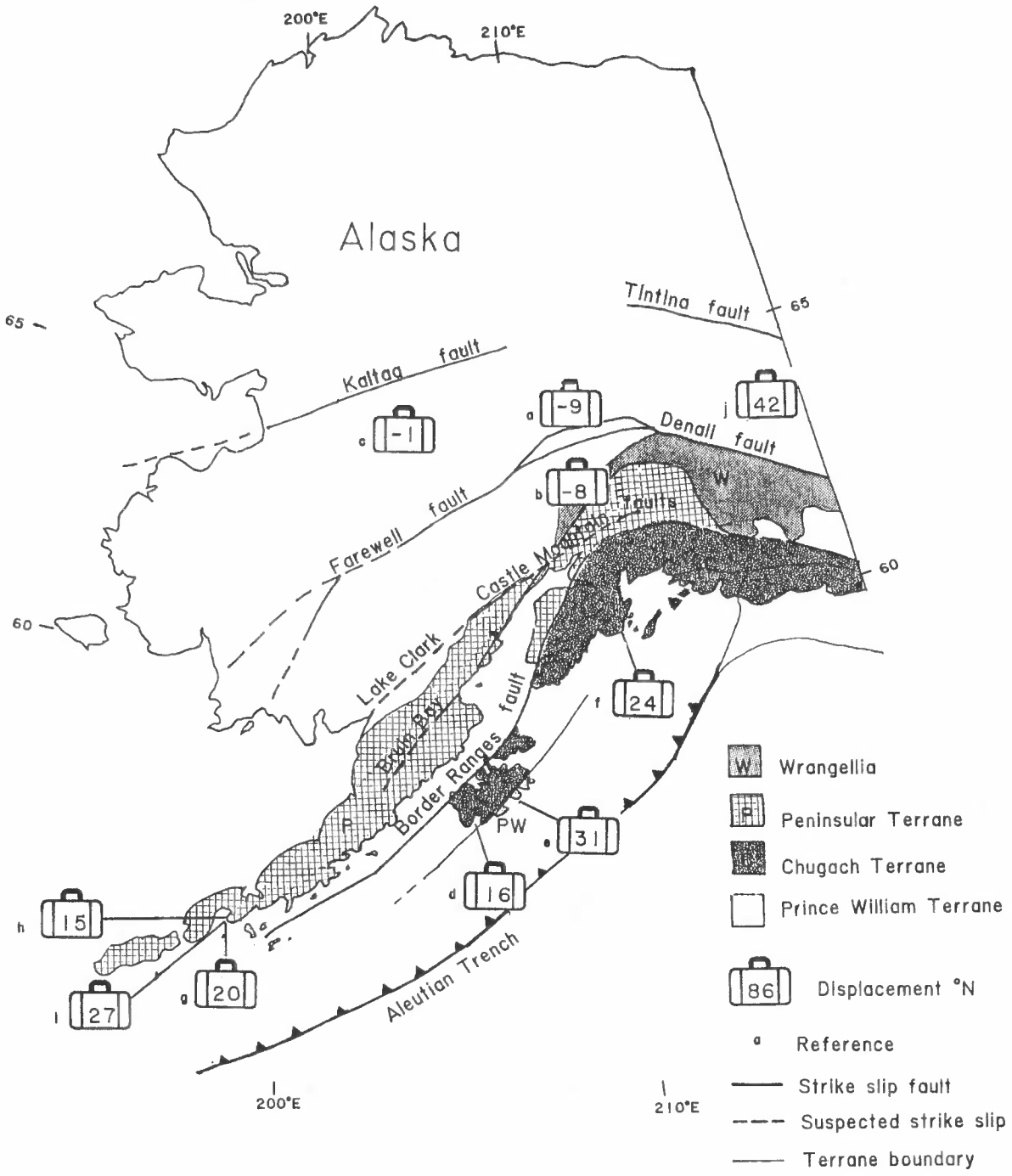


Fig. 1. Tectonostratigraphic terrane map of southern Alaska after Coe et al. (1985). Suitcases indicate location of paleomagnetic studies in latest Cretaceous to Paleocene/Eocene rocks. Numbers in suitcases indicate poleward displacement (+/- for northerly/southerly motion). a - Cantwell Fm, Hillhouse and Gromme (1982); b - Talkeetna Mts. unnamed lavas, Hillhouse et al. (1985); c - McGrath Notwina volcanics, Plumley (1984); d&e - Ghost Rocks volcanics, Plumley et al. (1983); f - Resurrection Peninsula sheeted dikes and pillow lavas, Gromme and Hillhouse (1981), f,h&i - Alaskan Peninsula sedimentary rocks, Stone et al. (1982), Stone and Packer (1977, 1979); j - MacColl Ridge, Panuska (1985).

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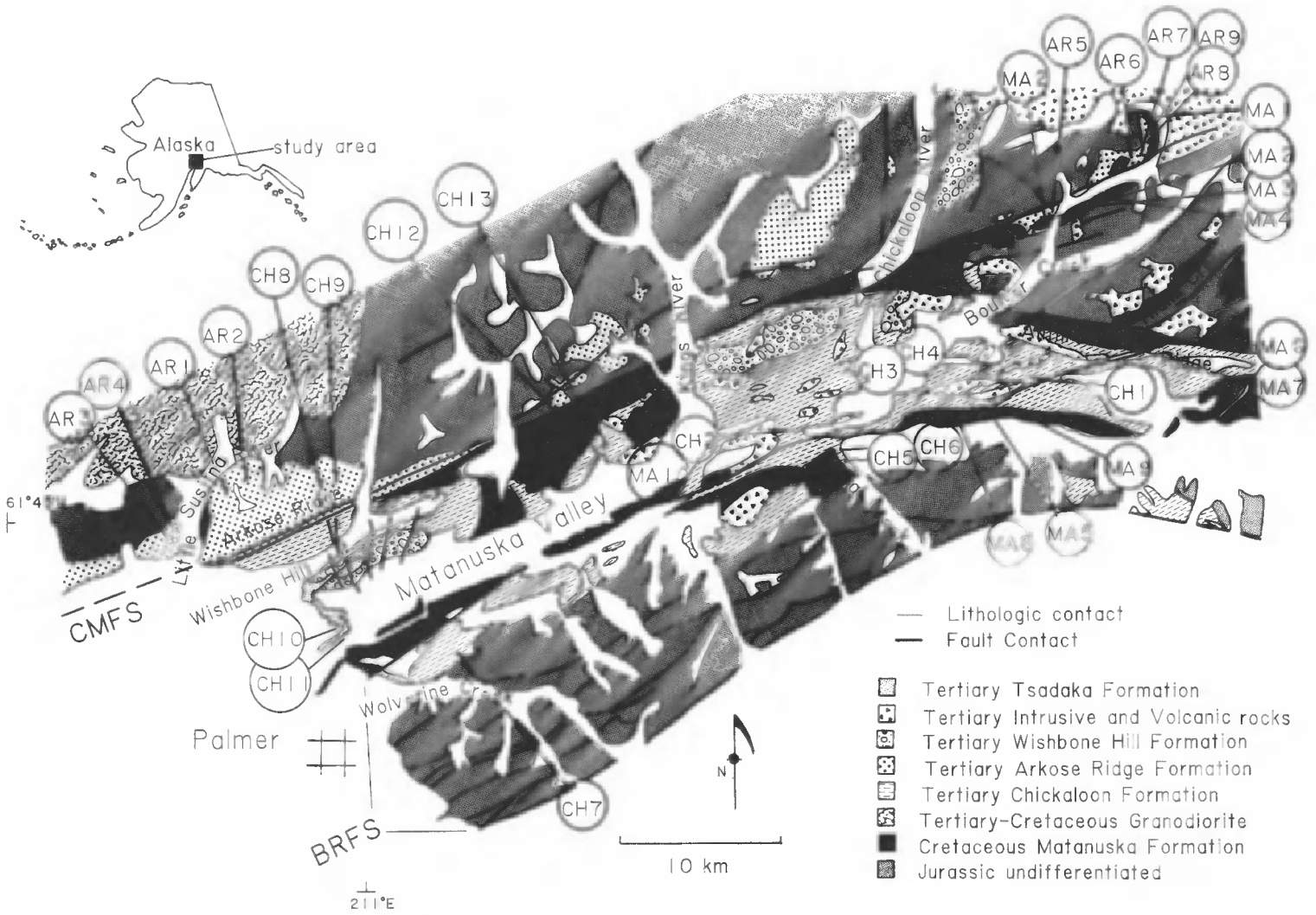


Fig. 2. Geologic map of the Matanuska Valley in southern Alaska [after G. R. Winkler, 1985, in written communication to T. L. Pavlis, and to L. E. Bruhn et al., in prep]. Geology of the Boulder creek area after Fuchs [1980]. CH - Chickaloon Formation sites, AR - Arkose Ridge Formation sites, MA - Matanuska Formation sites. CMFS - Castle Mountain Fault system, BRFS - Border Ranges Fault system.

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northeastern terminus of the Cook Inlet Basin in which the upper Mesozoic and Cenozoic rocks have been deformed and uplifted onto the flanks of the Chugach and Talkeetna mountains. Much of the Matanuska Valley is underlain by the Cretaceous Matanuska Formation and early Tertiary Chickaloon Formation which unconformably overlie older Mesozoic rocks [Martin and Katz, 1912; Barnes and Payne, 1956]. North of the Castle Mountain Fault, in the southern Talkeetna Mountains, Tertiary sediments of the Arkose Ridge Formation overlie both pre-Tertiary crystalline and sedimentary rocks [Martin and Katz, 1912; Barnes and Payne, 1956].

The middle and late Cretaceous sedimentation of the Matanuska Valley is characterized by a thick sequence of up to 3,050 m of marine siltstones and sandstones of the Matanuska Formation, deposited in what Payne [1955] has termed the Matanuska geosyncline. They are exposed throughout the Matanuska Valley, in the Nelchina area [Grantz, 1960] and have been penetrated in wells drilled in Cook Inlet Basin [Kirchner and Lyon, 1973]. In the western end of the Matanuska Valley, the Matanuska Formation unconformably overlies Upper Jurassic siltstones but further to the east, in the Nelchina area it unconformably overlies Lower Jurassic volcanic and associated sedimentary rocks [Jones and Grantz, 1967]. Jones and Grantz's [1963] revised lithogenetic column subdivides the Matanuska formation into 6 lithologic members bounded by intraformational unconformities. The longest of these unconformities spans almost 40 my, separating the lower three members (Lower Matanuska Formation) from the younger three members (Upper Matanuska Formation). Based on macrofossil evidence the Lower Matanuska Formation is Albian to Turonian in age [Imlay, 1959; Jones and Grantz, 1967], while the Upper Matanuska Formation is Campanian to Maestrichtian in age [Jones and Grantz, 1963]. In addition, member three has been further subdivided by Jones and Grantz [1963] into two characteristic fossil bearing sub-zones, the *Inoceramus schmidt* and *Pachydiscus kamishakensis* zones.

The Chickaloon Formation consists of over 1000 m of nonmarine and estuarine sediments of interbedded claystone, siltstone, quartzitic graywacke sandstone, feldspathic sandstone and conglomerate and includes numerous coal beds located in the upper half of the section [Barnes and Payne, 1956; Capps, 1927]. Megafossil floras from numerous sites in the upper part of the formation match floras from the Fort Union Formation and suggest a Paleocene age for the Chickaloon Formation [Wolfe et al., 1966]. More recent K-Ar and fission-track ages from volcanic ash partings in coal beds located in the Evans Jones Coal Mine range from 53.3 ±1.5 to 55.8 ±1.7 Ma [Triplehorn et al., 1984] indicating that the upper part of this formation straddles the Paleocene-Eocene boundary (55 Ma.).

Chickaloon rocks rest unconformably on the Matanuska Formation on Anthracite Ridge [Waring, 1936] and in the Wolverine Creek valley [Grantz and Wolfe, 1961], but are thought to lie conformably upon the Matanuska Formation near the mouth of Moose Creek and at Knob Creek [Barnes, 1962]. At Wishbone Hill, the Chickaloon Formation is overlain by the Wishbone Formation and although this contact is conformable in places [Barnes and Payne, 1956], local angular unconformities occur [Clardy, 1974]. In the eastern part of the valley the Cretaceous and early Tertiary sedimentary units are intruded by a series of both mafic and felsic dikes, stocks and large sills. Whole rock K-Ar analysis from five of these bodies yield ages between 37.5 and 45.2 ± 1.7 Ma [Silberman and Grantz, 1984].

The Arkose Ridge Formation consists of at least 500 m of non-marine well-indurated arkose, shale, sandstone and conglomerate which is characterized by clastic material derived from the Talkeetna Batholith [Barnes and Payne, 1956; Grantz and Wolfe, 1961; Detterman et al., 1974]. On Arkose Ridge, the sediments lie unconformably on the Talkeetna crystalline rocks and on pre-Jurassic mica schists [Grantz and Wolfe, 1961]. Further to the east, in Boulder Creek, Arkose Ridge Conglomerates lie conformably on top of the Matanuska Formation [Fuchs, 1980].

Age determinations for this formation are more poorly constrained than those of the Chickaloon Formation with estimates that range from Eocene [Martin and Katz, 1912] to Cretaceous [Barnes and Payne, 1956]. The conifers and angiosperms identified in the base of the section at Arkose Ridge were thought to be Albian or Cenomanian [Grantz and Wolf, 1961], and led these workers to conclude that the Arkose Ridge Formation was correlative with the lower Matanuska Formation. A pre-Tertiary age was also indicated by K-Ar age determination of 67.5 ±2.4 Ma made on a secondary biotite mineral from a graywacke [Csejtey et al., 1978]. However, the Arkose Ridge unconformably rests on a Lower to Middle Jurassic metamorphosed pelitic schist which yielded K-Ar dates that range from 59.0 ±1.8 to 65.0 ±3.0 Ma [Csejtey et al., 1977]. These results were interpreted to indicate that the schists underwent Paleocene retrograde metamorphism which presumably did not affect the overlying Arkose Ridge sediments. Radiometric ages from volcanic dikes tuffs, and flows intercalated in the Arkose Ridge sediments range from 45.5 ±1.8 to 56.2 ±1.7 Ma [Silberman and Grantz, 1984]. Yet, volcanic rocks in the southeastern Talkeetna Mountains, which are thought to lie stratigraphically above the Arkose Ridge sediments, yield similar K-Ar ages that range from 60.1 ±4.6 to 43.6 ±2.2 Ma [Silberman and Grantz, 1984]. More recent fossil evidence [Wolfe, 1981 in written communication to Silberman and Grantz, 1984] in which similar plant fossils were found in both the Arkose Ridge and Chickaloon formations suggest that these

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TABLE 1. Paleomagnetic Results From the Chickaloon Formation, Alaska

Site	Nc/N	S,D	M	k_1,k_2	α_{31},α_{32}	GEO INC	GEO DEC	STRAT INC	STRAT DEC	θ	ϕ	Φ
CH1	10/ 0	290,35										
CH2	18/09		P	-26.0,-1.1	9.0,10.2			59.8	342.2	66.3	66.7	49.3
CH2S	05	26,33	P	-75.4,-1.8	6.8,12.1	33.6	318.2	62.1	338.8			
CH2N	04	317,54	P	-40.0,-1.1	11.1,11.7	54.9	273.0	53.8	3.1			
CH3	18/ 0	90,67										
CH4	18/ 0	265,55										
CH5	20/06	80,71	P	-12.3,-1.7	14.7,24.7	-24.3	187.0	-73.3	281.4	45.0	257.2	31.0
CH6	20/ 0	265,57										
CH7	16/ 4		L	-127.5,-14.2	9.1,12.1	32.2	115.2	67.5	61.5	55.4	292.0	39.6
CH7	16/13		P	-14.9,-0.7	9.8,12.1			68.8	100.2	40.2	263.6	37.8
CH7A	5	235,60	P	-20.3,-2.0	12.0,20.0	10.5	130.2	66.1	106.7			
CH7B	4	89,80	P	-32.3,-2.1	10.1,19.1	21.5	23.4	63.9	118.1			
CH7C	4	226,75	P	-127.3,-1.6	5.0, 8.0	3.6	122.4	72.1	84.9			
CH8	17/12	58,60	P	-8.8,-0.3	14.4,15.9	43.3	358.7	62.9	95.6	35.7	272.7	45.7
CH9	17/11	273,95	P	-9.7,-0.4	14.1,16.0	-11.8	348.1	-67.2	226.3	62.3	305.3	40.1
CH10	14/10	184,22	L	-9.1,-6.3	11.7,14.4	-39.7	259.0	-60.8	249.7	45.2	295.0	49.1
CH10	14/12	184,22	P	-8.9,-0.2	15.1,16.2	-51.3	247.2	-68.7	222.3	65.5	305.5	38.0
CH11	14/ 0	89,50										
CH12	24/ 0	75,28										
CH13	14/06	250,38	P	-31.4,-1.3	9.1,12.1	51.3	126.9	69.4	58.8	58.4	290.1	36.9

Mean Paleomagnetic pole of 14 sample VGP's (see text): 47.9°(N), 299.1°(E), A_{95} = 12.6°. Mean direction of six site directions, Ch5, CH7, CH8, CH9, CH10, and CH13, derived from demagnetization plane analysis (see text): I_s = 70.3°, D_s = 73.9°, k_1 = −88.3, k_2 = −8.0, α_{31} = 3.1°, α_{32} = 9.3°. Mean paleomagnetic pole of six site VGP's derived from demagnetization plane data: 52.6°(N), 278.0°(E), A_{95} = 12.3°. Direction from the 60 individual demagnetization planes of sites CH5, CH7, CH8, CH9, CH10, and CH13: I_s = 69.8°, D_s = 76.8°, k_1 = −8.6, k_2 = −0.3, α_{31} = 6.6°, α_{32} = 7.3°. Paleomagnetic pole corresponding to this direction: 50.5°(N), 277.2°(E), δm = 12.18, δp = 7.7°.

NOTES: Upper hemisphere directions inverted through the origin when calculating site VGP. Site Coordinates: 211.8°(E), 61.5°(N). N_c = number of samples collected, N = number of samples used in analysis. S,D = strike and dip of beds. M = method of analysis, L = least square lines, P = remagnetization planes. k_1 and k_2 = Bingham concentration parameters Onstott [1980]. α_{31} and α_{32} = Bingham 95% confidence limits [Onstott, 1980]. θ and ϕ = latitude and east longitude of virtual geomagnetic pole. Φ = paleomagnetic colatitude. A_{95} = 95% confidence limit about mean pole. I_s and D_s = inclination and declination of tilt-corrected direction. δm,δp = 95% confidence oval about pole position.

units are coeval and that the Arkose Ridge Formation is Paleocene to Eocene in age.

The structural history of the region is complex and includes multiple deformation events. In addition to the uplift and erosion evidenced by the intraformational unconformities in the Matanuska Formation [Jones and Grantz, 1967], the rocks of the Matanuska Valley and flanking mountains experienced a major period of folding, uplift, and erosion between the middle Eocene and late Oligocene [Barnes and Payne, 1952; Barnes, 1962]. Fold axes trend east-northeast with shallow plunges. The upper Oligocene to lower Miocene(?) Tsadaka Formation that rests unconformably on the older folded rocks has only been gently warped [Clardy, 1975]. Neogene deformation has been confined to movement on three fault sets, dip-slip and dextral strike-slip motion on steeply dipping east-trending faults including the Castle Mountain Fault, dextral motion on west-northwest trending strike-slip faults, and sinistral movement on northeast trending strike-slip faults [Bruhn and Pavlis, 1981]. Paleomagnetic evidence from the silicic and mafic sills that intrude the Chickaloon Formation [Stamatakos et al., 1988] indicate that the region between the Castle Mountain Fault and the Border Ranges Fault has undergone clockwise rigid-block rotations due to differential right-lateral shear on these bounding faults that may be related to the Neogene deformation. Quaternary right-lateral motion on the Castle Mountain Fault has been documented by offset glacial deposits [Detterman et al., 1974; Fuchs, 1980] and recent seismic activity [Lahr et al., 1986].

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Paleomagnetic Sampling and Laboratory Methods

All the samples for this study except for samples from sites AR7 and AR9 were drilled in the field using a portable gasoline-powered drill fitted with a 2.5 cm diameter diamond-chip coring bit. Each sample was oriented with a Pomeroy Industries Unlimited core orientation device and magnetic compass. Because of steep and unstable slopes, samples from sites AR7 and AR9 were obtained from oriented hand samples collected in Boulder Creek. In the Chickaloon Formation, 217 samples were taken from 13 sites, and from the Arkose Ridge Formation, 111 samples were taken at 9 sites (Figure 2, Tables 1 and 2). In the Matanuska Formation, 174 samples were collected from 10 sites (Figure 2, Table 3). At each site, the attitude of bedding was measured on several bedding surfaces and averaged, giving an error estimate for these measurements of approximately 2°. In order to constrain the age and determine the nature of the magnetizations in these rocks, sites were chosen to offer a variety of stability tests that included a conglomerate test (site AR2) in which samples were collected from granitic cobbles in a basal conglomerate bed, several fold tests (sites CH2, CH7) in which samples were collected from both limbs of outcrop scale folds, and a baked-contact test (site MA10) in which an evenly-spaced sequence of samples were collected from the center of a small dike into the hornfels equivalent of the Matanuska shale country rock.

The Chickaloon sites were distributed over 25 km along strike and stratigraphically range from the base of the section at sites CH7 (near Wolverine Creek), CH5 and CH6 (located near Coal Creek on the south side of the Matanuska River), to the upper part of the section at sites CH8, CH9, CH12 CH13 (in the coal bearing units at the Premier, Jones and Eska coal mines) (Figure 2). The Arkose Ridge sites were divided between Boulder Creek, to the east, and Arkose Ridge, to the west. Sites AR1 and AR2 were located at the base of the Arkose Ridge section directly on Arkose Ridge. Sites AR3 were located along the Little Susitna River, just north of the Castle Mountain Fault. Site AR5 was located at the base of the Arkose Ridge section in Boulder Creek, just up-section from the Matanuska site MA1. The other Arkose Ridge sites, AR6 to AR9 were collected on a traverse of the section along a tributary of Boulder Creek. Because of local structural complications it was impossible to estimate the total stratigraphic coverage of the Arkose Ridge Formation. However, a minimum estimate, based on sites AR6 to AR9, is approximately 400 m of section from the bottom of the tributary valley to its source (Figure 2).

All of the Matanuska sites are from the Upper Matanuska Formation. Site MA1 was located near the contact with the Arkose Ridge Formation in the upper reaches of Boulder Creek. Sites MA2, MA3 and MA4, south of the Caribou Splay of the Castle Mountain Fault, were located in the *Pachydiscus kamishakensis* zone of Jones and

TABLE 2. Paleomagnetic Results From the Arkose Ridge Formation, Alaska

Site	Nc/N	S,D	M	k_1,k_2	α_{31},α_{32}	GEO INC	GEO DEC	STRAT INC	STRAT DEC	θ	ϕ	Φ
AR1	15/11	42,64	P	-6.8,-0.2	17.9,19.4	10.5	310.2	73.9	306.7	64.8	141.8	30.0
AR2	10/10	101,53	L	-3.2,-1.2	32.3,69.1	-10.1	158.1	-49.3	135.7			
AR3	11/09	70,99	L	-32.8,-3.4	6.0,20.6	-73.7	344.8	25.2	340.8	39.4	56.9	76.8
AR3	11/09	70,99	P	-12.2,-3.5	11.3,31.5	-52.2	329.2	46.1	330.3	50.4	75.3	62.5
AR4	9/09	74,64	L	-30.6,-17.5	6.0,20.6	6.5	144.5	-52.4	131.1	47.8	101.9	51.9
AR4	9/09	74,64	P	-52.1,-1.4	5.9, 7.8	3.5	149.7	-57.4	136.8	54.5	100.0	57.0
AR5	8/ 0	338,32										
AR6	15/12	326,38	L	-10.9,-8.2	9.5,11.0	-40.0	95.3	-60.9	138.3	58.4	102.4	48.1
AR6	15/11	326,38	P	-8.9,-0.7	12.8,16.2	-35.7	79.5	-66.7	110.8	51.0	140.0	40.7
AR7	13/ 0	50,41										
AR8	18/13	40,46	L	-10.3,-0.9	11.8,17.3	25.8	307.6	71.7	302.0	60.8	137.9	33.5
AR8	18/11	40,46	P	-8.7,-4.5	6.2, 9.0	33.2	306.5	78.8	294.7	63.0	163.0	21.6
AR9	12/ 0	348,20										

Mean paleomagnetic pole of sites AR3, AR4, AR6, and AR8 using line data (see text): 55.2°(N), 95.1°(E), A_{95} = 13.2°. Mean direction of the four site directions, AR1, AR4, AR6, and AR8, derived from demagnetization plane analysis (see text): I_s = 69.6°, D_s = 304.35°, k_1 = -145.7, k_2 = -24.1, α_{31} = 4.2°, α_{32} = 10.3°. Mean paleomagnetic pole of four site VGP's derived from demagnetization plane data: 60.4°(N), 134.6°(E), A_{95} = 13.0°. Direction from the 42 individual demagnetization planes of sites AR1, AR4, AR6, and AR8: I_s = 71.5°, D_s = 301.7°, k_1 = - 12.0, k_2= -0.2, α_{31} = 7.2°, α_{32} = 7.6°. Paleomagnetic pole corresponding to this direction: 60.2°(N), 138.6°(E), δm = 11.6°, δp = 6.4°.

NOTES: Site AR2 - Conglomerate test, see text for discussion. For symbol definitions see table 1.

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TABLE 3. Paleomagnetic Results from the Matanuska Formation, Alaska

Site	Nc/N	S,D	T		Geo			Strat				
				k_g	Inc	Dec	k_s	Inc	Dec	θ	ϕ	Φ
MA1	20/10	125,05	530 °C	18	-67	106	18	-65	96	46	144	43
MA2	9/06	262,11	530 °C	34	-56	63	34	-58	79	29	151	51
MA3	15/11	292,25	530 °C	101	-73	100	101	-62	163	69	68	47
MA4	20/09	262.15	530 °C	23	-63	94	23	-57	118	46	119	52
MA5	15/ 0											
MA6	16/07	247,58	350 °C	12	71	39	12	22	355			
MA7	18/10	289,81	450 °C	15	2	178	15	66	128	28	248	42
MA8	17/10	317,72	450 °C	18	-26	56	18	-78	180	85	212	23
MA9	19/13	245,71	350 °C	123	68	7	123	0	346			
MA10	25/11	45,49	450 °C	39	50	331	39	75	90	51	260	28

Mean paleomagnetic pole of sites MA1, MA2, MA3, and MA4: 50°(N), 131°(E), A_{95} = 22.5°. Mean directions of sites MA7, MA8, and MA10, south of Castle Mountain fault: I_g = 42°, D_g = 228°, k_g = 2, I_s = 78°, D_s = 94°, k_s = 22. Mean direction of 20 samples from sites Ma6 and Ma9: I_g = 70°, D_g = 21°, k_g = 107, I_s = 11°, D_s = 350°, k_s = 25.

Notes: Upper hemisphere directions inverted through the origin when calculating site VGP. Site coordinates: 211.8°(E), 61.5°(N). N_c = Number of samples collected and demagnetized, N = number used in analysis. S,D = Strike and dip of beds. T = maximum unblocking temperature for magnetic component. k = Fisher precision parameter (k_s = stratigraphic k, k_g = geographic k). θ and ϕ = latitude and east longitude of virtual geomagnetic pole. Φ = paleomagnetic colatitude. A_{95} = 95% confidence limit for mean pole. I_g and D_g = inclination and declination without corrections for tilt. I_s and D_s = inclination and declination of tilt-corrected directions.

Grantz [1963] within 750 to 1000 m of the top of the Matanuska section. South of Castle Mountain Fault 6 Matanuska sites, MA5 to MA10, were distributed over 35 km along strike from just east of Hicks Creek to just west of the Chickaloon River. The two Hicks Creek sites, MA6 and MA7, were located in the *Inoceramus schmidt* zone of Jones and Grantz [1963] approximately 1500 m from the top of the section. The other Matanuska sites in the Matanuska Valley, south of the Castle Mountain Fault, were are also located in the *Pachydiscus kamishakensis* zone of Jones and Grantz [1963] (Figure 2).

Progressive thermal demagnetization (12-20 steps) was performed on 118 of the 217 Chickaloon samples, 72 of the 111 Arkose Ridge samples, and all 174 Matanuska samples. The remaining 99 Chickaloon samples and 39 Arkose Ridge samples were progressively demagnetized (10-15 steps) in peak alternating fields (af) up to 80 mT. In addition, 38 of the Chickaloon samples, 13 of the Arkose Ridge samples, and 6 of the Matanuska samples were long enough to yield companion specimens which were also progressively demagnetized (10-15 steps) in peak alternating fields up to 80 mT for comparison with the thermal results. All the thermal demagnetization was conducted with a Schonstedt TSD-1 thermal demagnetizer. Alternating field demagnetization of the Chickaloon and Arkose Ridge samples was performed on Stanford University's af demagnetizer. Alternating field demagnetization of the Matanuska samples was conducted on a Schonstedt GSD-5 tumbling AC demagnetizer. Remanence measurements for the thermally demagnetized Chickaloon, Arkose Ridge samples, and all the Matanuska samples were made on a Molspin spinner magnetometer, which is situated in a coil-controlled low-field environment to enhance its sensitivity. Remanence measurements for the af demagnetized Chickaloon and Arkose Ridge samples were made on an SCT two-axis cryogenic magnetometer at Stanford University. Susceptibility measurements were made on a Sapphire Instruments SI-2 susceptibility bridge.

Characteristic magnetizations were determined from principal component analysis [Kirschvink, 1980]. Where multiple components of magnetization with overlapping coercivity spectrum were encountered, the least-squares remagnetization plane analysis [Halls, 1976; Kirschvink, 1980] was applied. When a sample was long enough to yield two companion specimens, and both specimens contained stable magnetic directions, they were averaged to a sample direction. Site-mean directions were then calculated from the sample directions, assigned 95% confidence intervals both as Fisher distributions [Fisher, 1953] and as Bingham distributions [Onstott, 1980] and then converted to virtual geomagnetic poles (VGP). Where possible, the statistical significance of fold

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test and regional tilt tests was determined following the method suggested by McFadden and Jones [1981], otherwise the method of McElhinny [1964] was applied.

Magnetic carriers in these rocks were characterized by anhysteretic remanent magnetization (ARM), in which samples were given an ARM (DC field = 0.04 mT, peak AF = 100 mT) and were then stepwise demagnetized in an alternating field up to 100 mT, and by isothermal remanent magnetization (IRM) acquisition experiments, in which samples were given an IRM in a DC field of 150 mT and then af demagnetized. The bulk susceptibility and anisotropy of magnetic susceptibility (AMS) were also measured to estimate the magnetic grain size [King et al., 1982] and the orientation and magnitude of the magnetic fabric. In addition, several samples were subjected to progressively higher DC magnetic fields from 10 to 500 mT and then thermally demagnetized to a maximum temperature between 585 °C and 685 °C. In order to determine whether new magnetic phases were produced during thermal demagnetization, these samples were then remagnetized in a DC field of 200 mT and their remanence remeasured.

In a series of redeposition and recompaction experiments, two representative samples from the Arkose Ridge Formation were disaggregated in distilled water by continuous vibration in a sonicator for a period of approximately two weeks. These disaggregated sediments were then inspected under an optical microscope (100x) and analyzed on a Phillips model AP-3600 powered x-ray diffractometer. Four samples were then redeposited in a acrylic cylinder (1.37 cm ID, 4.4 cm. long) in distilled water in the earth's field (I = 60°) over a period of 30 days. These redeposited sediments were then compacted in a consolidometer [Anson and Kodama, 1987], situated inside a set of 1.0 m Helmholtz coils adjusted to generate a 60° inclination field. These samples were consolidated by a uniaxial pressure which increased from 0.00 MPa to a maximum pressure of 0.30 MPa over approximately twelve hours. During compaction, the remanence of the samples was periodically measured on a two-axis CTF cryogenic rock magnetometer. After compaction, the samples were demagnetized in an alternating fields up to 100 mT using a Schonstedt GSD-5 tumbling demagnetizer. In addition, seven slurries with water contents in excess of 100% by weight [(%water = (weight wet - weight dry)/weight dry)], made from distilled water and the disaggregated sediment, were given magnetizations by stirring them in magnetic fields generated by the Helmholtz coils. The samples were subsequently compacted in pressures up to 0.30 MPa and then af demagnetized. Stirring was used to impart an analogue of postdepositional remanent magnetization [Tucker, 1980] which would be parallel to the ambient field. Helmholtz coils were used to offset a strong vertical field (75,000 gamma) located above the access port of the magnetometer. These coils were in place for the stirred slurry/compacting experiments but not for the redeposition/compaction experiments.

Paleomagnetic Results

Chickaloon Formation

Natural remanent magnetizations (NRM) of the Chickaloon samples were of both normal and reversed polarity with highly scattered directions and weak intensities (between 0.05 and 5.0 mA/m). During thermal demagnetization, these samples exhibited distributed unblocking temperatures up to 400-450 °C above which many of the samples showed a rise in magnetic intensity. Only samples from sites CH7 and CH10 could be thermally cleaned to 575 °C (Figure 3). During af demagnetization, about 20% of the samples exhibited unstable magnetic directions. The remaining 80% of the af demagnetized samples showed more easily interpretable linear demagnetization trends having moderate coercivities with mean destructive fields (MDF's) between 15 and 30 mT except for samples from site CH10, which have higher coercivities (MDF's > 40 mT).

Orthogonal projections of demagnetization data [Zijderveld, 1967] revealed that many of these samples contained multiple component magnetizations with curved demagnetization trajectories suggesting overlapping coercivity and unblocking spectra (Figure 4). Of the 255 Chickaloon specimens demagnetized, characteristic magnetizations were isolated for only fourteen samples (defined by a minimum of four demagnetization points and the origin (MAD < 15%). Four of these samples, those from site CH7, showed a very weak intensity (0.5 mA/m) and high unblocking temperature component (420-560 °C) with moderately shallow and southwesterly directions in geographic coordinates and steeply down and northeasterly directions in stratigraphic coordinates. The remaining ten directions are from both af and thermal demagnetizations of samples from site CH10. These magnetizations were up and to the southwest with shallow inclinations in geographic coordinates. Upon correction for bedding tilt, these upward, southwesterly directions become steeper and antipodal to the directions from site CH7. Results from these line data for these two sites are summarized in Table 1. In addition, eleven samples from site CH7 yielded an intermediate magnetization that was unblocked between 100 °C and 420 °C. The direction of this component is easterly and steeply down in geographic coordinates. However, when these samples are corrected for bedding tilt, the directions diverge, failing the fold test [Graham, 1949].

Demagnetization plane analysis in which planes were defined by at least 5 demagnetization points above 15 mT or unblocking temperatures in excess

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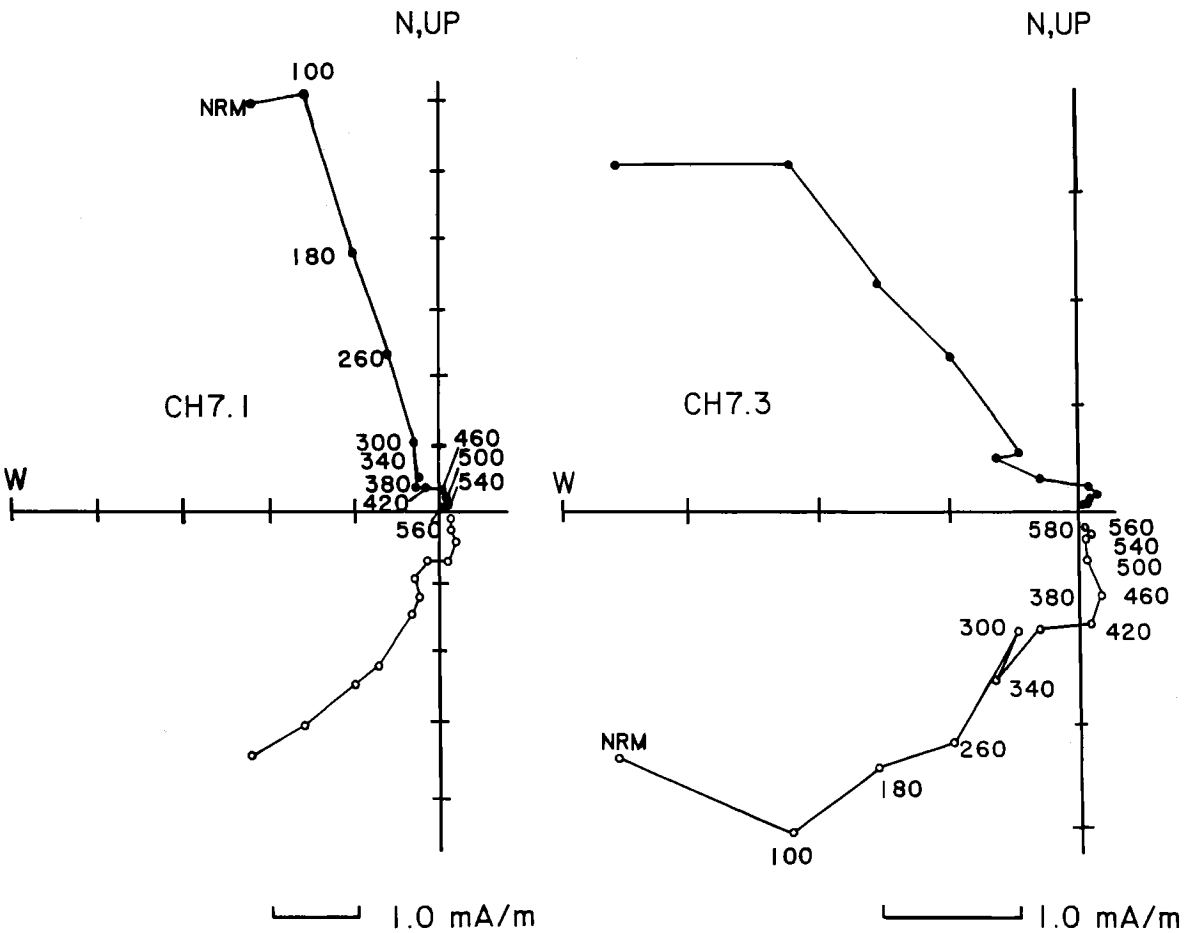


Fig. 3. Orthogonal projection of thermal demagnetization data for samples from Chickaloon Formation site CH7. Note that Characteristic magnetizations are isolated only at highest unblocking temperatures. Open circles - vertical component; solid circles - horizontal component. Demagnetization steps in °C.

of 367 °C, yielded 69 demagnetization planes from 7 of the 13 sites. In geographic coordinates, the planes do not appear to converge (Figure 5), but in stratigraphic coordinates, the planes converge on a northeast and moderately to steeply down direction.

In addition, the demagnetization planes were used to define individual site directions (Table 1, Figure 6). Demagnetization plane directions for sites CH7 and CH10 are statistically indistinguishable from the corresponding line data (Figure 7). Two sites, CH2 and CH7, had outcrop scale folds with samples from opposite fold limbs. For both sites, if the magnetization direction for each limb is derived from the demagnetization planes for that limb, the limb directions converge when the beds are restored to their horizontal position (Figure 8). Because there are only two limb directions for the fold at CH2 the statistical significance of this fold test could not be evaluated. At site CH7, the limb directions pass the fold test at the 95% confidence level [McElhinny, 1964]. Furthermore, if the demagnetization planes are used to derive individual site directions, and these seven directions are corrected for bedding tilt, all but site CH2 converged on an east-northeasterly and steeply down direction or a west-southwesterly and steeply up direction (Figure 9). These 7 site directions pass the fold test at the 95% confidence level [McElhinny, 1964].

Arkose Ridge Formation

Natural remanent magnetizations from the Arkose Ridge samples were also of both polarities with scattered directions and moderate to weak intensities (between 1.0 and 10.0 mA/m). During thermal demagnetization, these samples exhibited a narrower range of unblocking temperatures than

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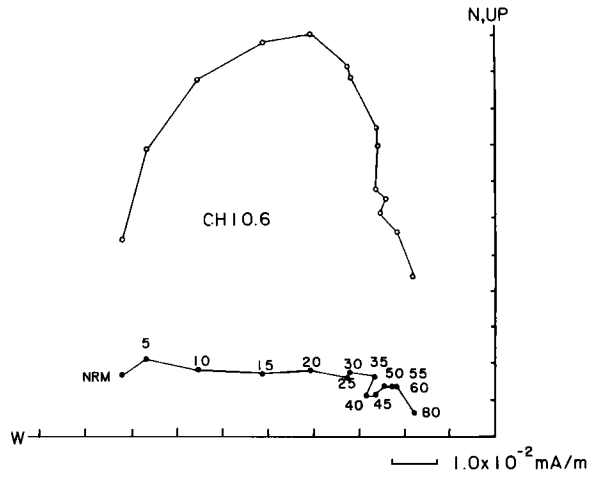


Fig. 4. Orthogonal projection of af demagnetization data from Chickaloon Formation site CH10. Note curved trajectory indicating overlapping coercivity spectra of primary and secondary magnetizations. Demagnetization steps in mT. Symbols same as in figure 3.

the Chickaloon rocks, between 500 °C and 575 °C (Figure 10). Although several samples showed an increase in magnetic intensity above the 367 °C step, this behavior was not as pervasive as in the Chickaloon samples. Af demagnetization revealed moderate to high coercivities (MDF's between 30 and 50 mT) and like the Chickaloon samples, about 20% of these samples showed nonsystematic behavior during af demagnetization which made interpretations impossible.

Samples from sites AR5, AR7, and AR9 did not yield any consistent demagnetization trends during either af or thermal demagnetization. Samples from site AR1 yielded only demagnetization planes. Characteristic magnetizations were isolated in samples from site AR2, the basal conglomerate, which were randomly distributed in both geographic and stratigraphic coordinates (precision parameter (K) in geographic = 1.06, K in stratigraphic = 1.10). Characteristic magnetizations were also isolated from both af and thermally demagnetized samples from the remaining Arkose Ridge sites (AR3, AR4, AR6, and AR8). Upon correction for bedding tilt these directions converge on a northwest and moderate to steeply down direction or a nearly antipodal southeast and moderate to steeply up direction. For these four site directions, the hypothesis of a common true mean direction was rejected at the 95% confidence level when the in-situ directions were used and accepted when the tilt-corrected directions were used [McFadden and Jones, 1981].

Remagnetization plane analyses yielded 51 demagnetization planes from 5 sites (Table 2). Similar to the Chickaloon rocks, the poles to these planes appear to be randomly distributed in

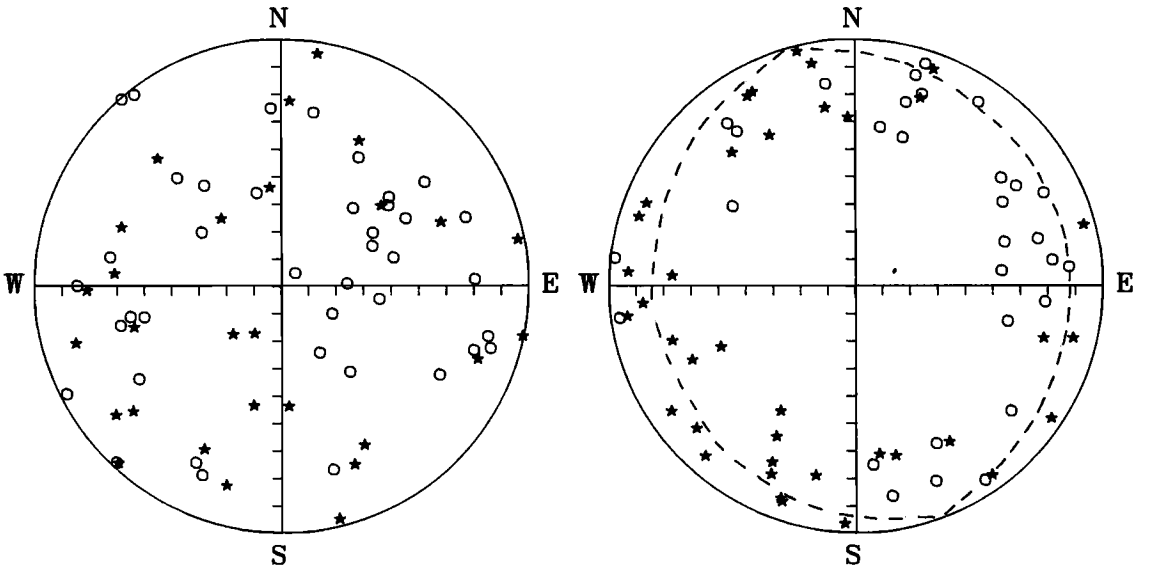


Fig. 5. Stereonets showing poles to demagnetization planes for Chickaloon site CH2, CH5, CH7, CH8, CH9, CH10 and CH13 in both geographic (Geo) and stratigraphic (Strat) coordinates. Stars - lower hemisphere poles; open circles - upper hemisphere poles; Dashed line - best fit great circle of poles to demagnetization planes.

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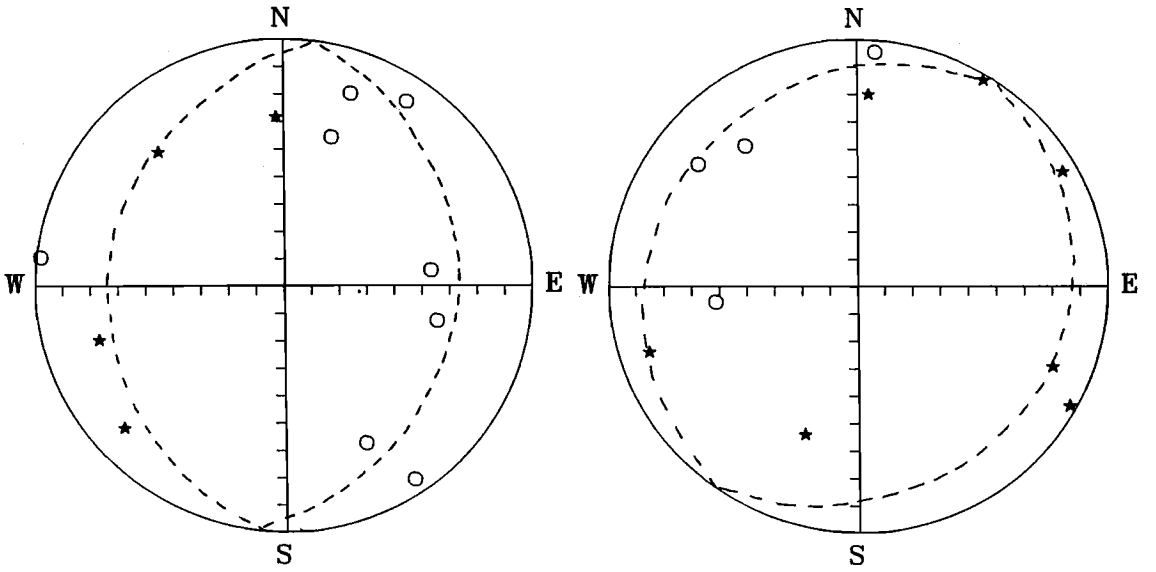


Fig. 6. Stereonets showing poles to demagnetization planes for Chickaloon Formation site CH8 and Arkose Ridge Formation site AR2, both in stratigraphic coordinates. Symbols same as in figure 5.

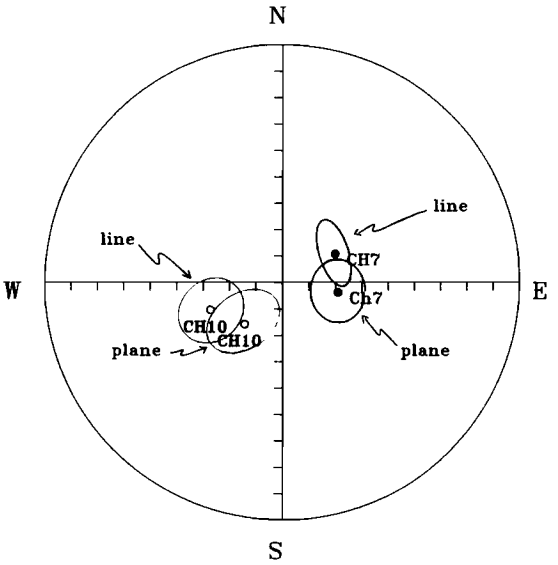


Fig. 7. Stereonet showing, in stratigraphic coordinates, site mean directions (line) from characteristic magnetizations and site directions (plane) from intersecting demagnetization planes for Chickaloon Formation. Note that characteristic magnetizations and demagnetization plane data yield similar directions. Open symbols - upper hemisphere; solid symbols - lower hemisphere; shaded region - 95% Bingham confidence ovals.

geographic coordinates but, in stratigraphic coordinates, they appear to lie on a great circle. The pole to this great circle has a northwest and moderate to steeply down direction which is steeper than the line derived Arkose Ridge directions (Figure 11).

If the demagnetiztion analysis is applied to each site individually then the demagnetization plane directions appear to be slightly steeper than the line derived directions obtained from the characteristic magnetizations, however, the the 95% confidence limits for the corresponding sets of directions all overlap (Figure 12, Table 2). These five demagnetization plane-derived site directions pass the fold test at the 95% confidence level [McElhinny, 1964] (Figure 13). In addition, the demagnetization plane-derived site directions exhibit greater antiparallelism of the reversed and normal directions than the site directions derived from the characteristic magnetizations.

Matanuska Formation

Pilot af demagnetizations of six Matanuska samples were ineffective in demagnetizing these rocks and thus, all the remaining Matanuska samples were thermally demagnetized. The magnetizations and demagnetization behavior of the Matanuska rocks were decidedly different depending on whether they were from north or south of the Castle Mountain Fault. North of the fault (sites MA1 to MA4), initial NRM intensities

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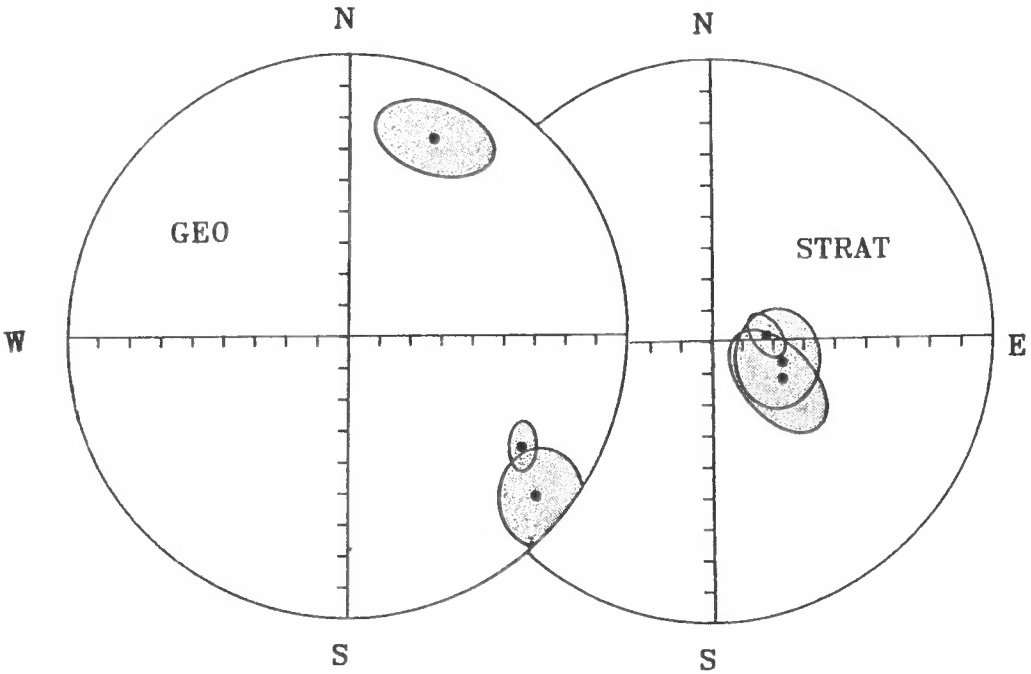


Fig. 8. Stereonets showing directions derived from demagnetization planes from three different limbs orientations of fold at Chickaloon site CH7 in geographic (Geo) and stratigraphic coordinates (Strat). Note passage of local fold test. Symbols same as in figure 7.

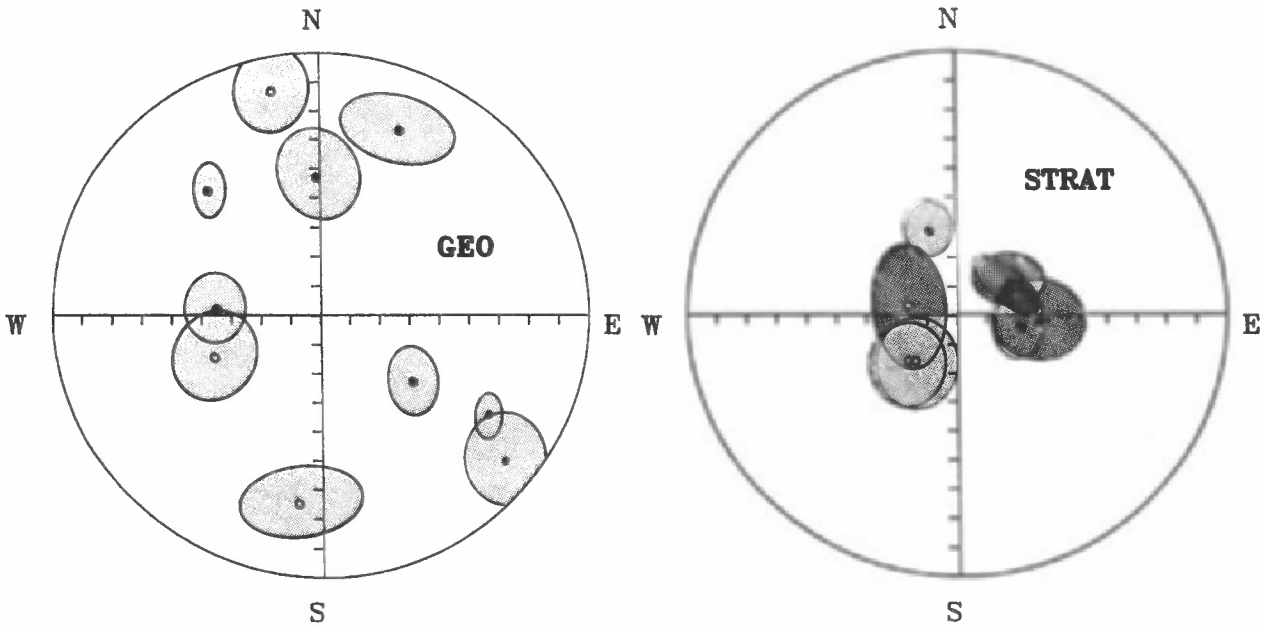


Fig. 9. Stereonets showing directions derived from intersecting demagnetization planes in geographic (Geo) and stratigraphic (Strat) coordinates for Chickaloon Formation. Note antipodal directions and passage of a regional tilt test. Symbols same as in figure 7. Dark shaded oval in right stereonet (Strat) is mean of the site directions with its associated 95% Bingham confidence oval.

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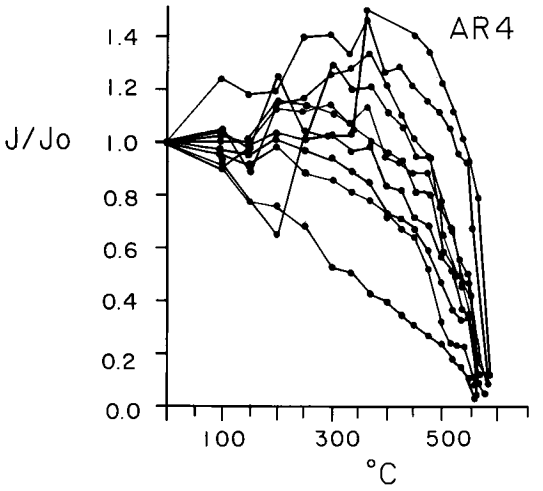


Fig 10. Normalized intensity plot of thermal demagnetization data from Arkose Ridge Formation. Note narrow range of unblocking temperatures.

ranged from 1.0 and 10.0 mA/m and the NRM directions were scattered. These samples had distributed unblocking temperatures up to at least 500 °C above which the magnetic intensities increased rapidly with increasing demagnetization temperature. Principal component analysis revealed two magnetizations, a low unblocking temperature (<300 °C) component that is oriented steeply down and a high unblocking temperature component (≤525 °C) that is southwesterly and moderately to steeply up. Because the bedding tilt for these sites is small, less than 11°, the directions are similar in both stratigraphic and geographic coordinates. Demagnetization plane analyses for these same four sites yield statistically indistinguishable directions at the 95% confidence level. Tilt correcting the four in-situ directions reduced the Fisher precision parameter (K) from 53.0 to 22.5 (Figure 14). However, using either data set, the hypothesis of a common true mean direction was accepted at the 95% confidence level, when either the stratigraphic or geographic coordinates were used [McFadden and Jones, 1981].

South of the Castle Mountain fault, the Matanuska magnetizations were characterized by low NRM intensities (0.3 to 1.0 mA/m) and nonsystematic demagnetization behavior. For samples from sites MA5, MA6 and MA9 magnetizations were unblocked by 300 °C. Characteristic magnetizations isolated from 20 samples from sites MA6 and MA9 are steeply downward and northerly and coincide with the present day field direction in geographic coordinates (Figure 15). When the directions from these two sites are corrected for tilt, the directions become quite shallow and northerly.

Demagnetization of samples from sites MA7 and MA8 revealed two magnetization components, one with low unblocking temperatures (150 °C to 300 °C) that has an in-situ direction that is similar in direction to the low unblocking component of site MA6 and MA9, and one with unblocking

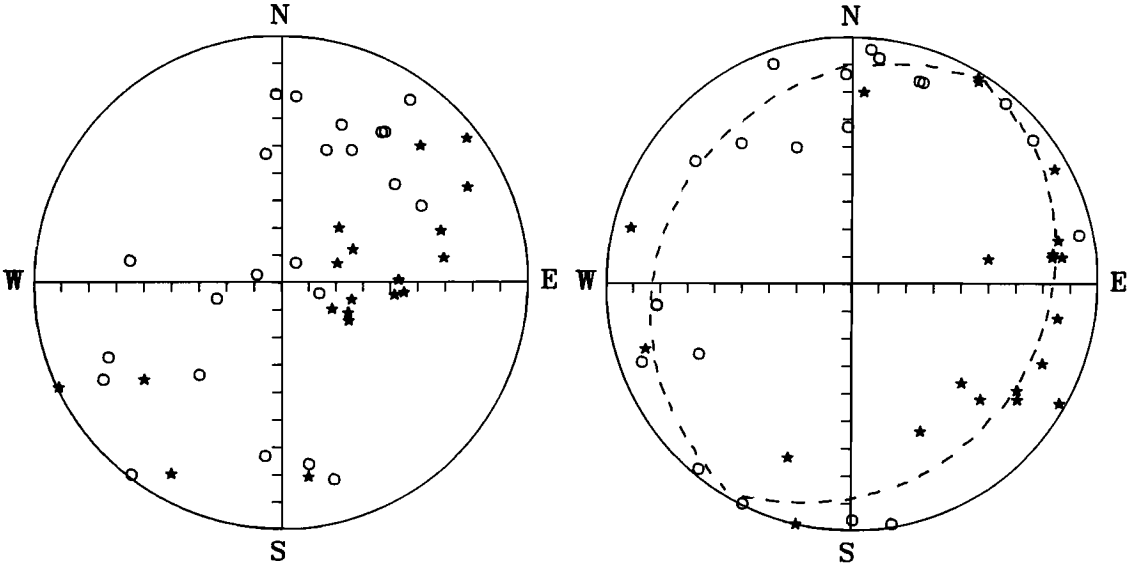


Fig. 11. Stereonets showing poles to demagnetization planes for Arkose Ridge sites AR1, AR3, AR4, AR6, and AR810 in both geographic (Geo) and stratigraphic (strat) coordinates. Symbols same as in figure 5.

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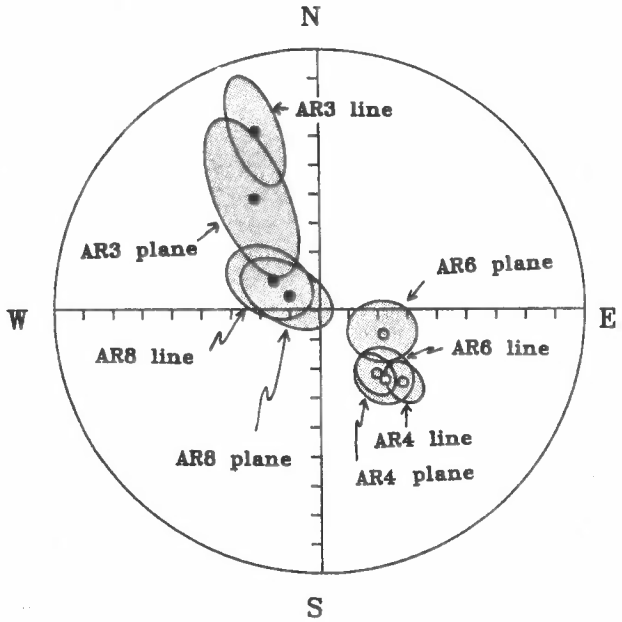


Fig. 12. Stereonet showing, in stratigraphic coordinates, site mean directions (line) from characteristic magnetizations and site directions (plane) from intersection demagnetization planes. Symbols same as in figure 7. Note that although two methods yield site directions that have overlapping 95% Bingham confidence ovals, plane-derived directions are steeper than line-derived directions.

temperatures between 333 °C and 450 °C that has a very steep tilt-corrected direction.

The hornfels at site MA10 had NRM intensities (5.0 to 70.0 mA/m) much higher than typical Matanuska samples. Demagnetizations produced linear trajectories and allowed isolation of characteristic magnetizations which were moderately down to the northwest in geographic coordinates and steeply down and to the east in stratigraphic coordinates. Igneous samples from the dike at this site were very weakly magnetized (1.0 to 5.0 mA/m) with low coercivities (MDF = 8 mT) and low unblocking temperatures (<200 °C). No directional results could be obtained from these samples. Using the hornfels direction from site MA10 and the two high unblocking temperature directions from sites MA7 and MA8 the hypothesis of a common true mean direction is rejected when the in-situ directions are used and accepted when the tilt corrected direction are used [McFadden and Jones, 1981] (Figure 16).

Rock Magnetic Results

ARM and Susceptibility

In the granulometry experiments, the ratios of ARM to bulk susceptibility for the Chickaloon and Arkose Ridge samples were generally high, ranging from 0.8 to 10.0. Only site AR8 had a relatively low ratio of 0.1. Anisotropy of magnetic susceptibility (AMS) ellipsoids had no consistent orientations, and did not have the minimum axes oriented perpendicular to bedding. The percent

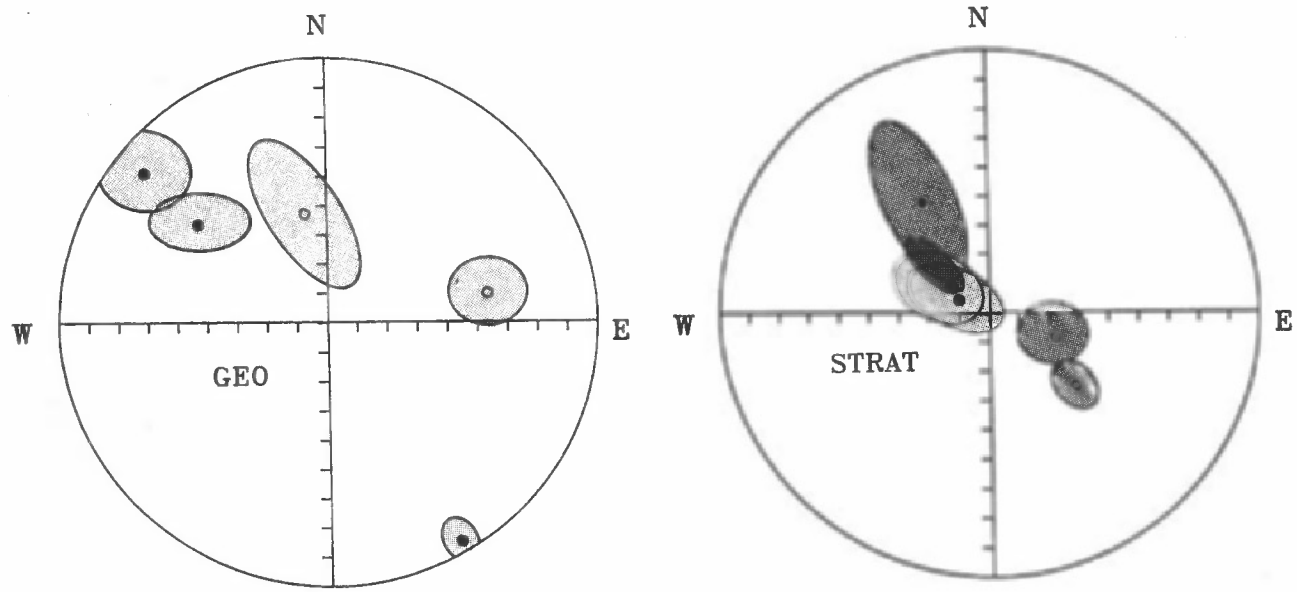


Fig. 13. Stereonets showing site directions derived from intersecting demagnetization planes in geographic (Geo) and stratigraphic coordinates for Arkose Ridge Formation. Note passage of fold test and antipodal directions. Symbols same as in figure 7. Dark shaded oval in right stereonet (Strat) is the mean of site directions with its associated 95% Bingham confidence oval.

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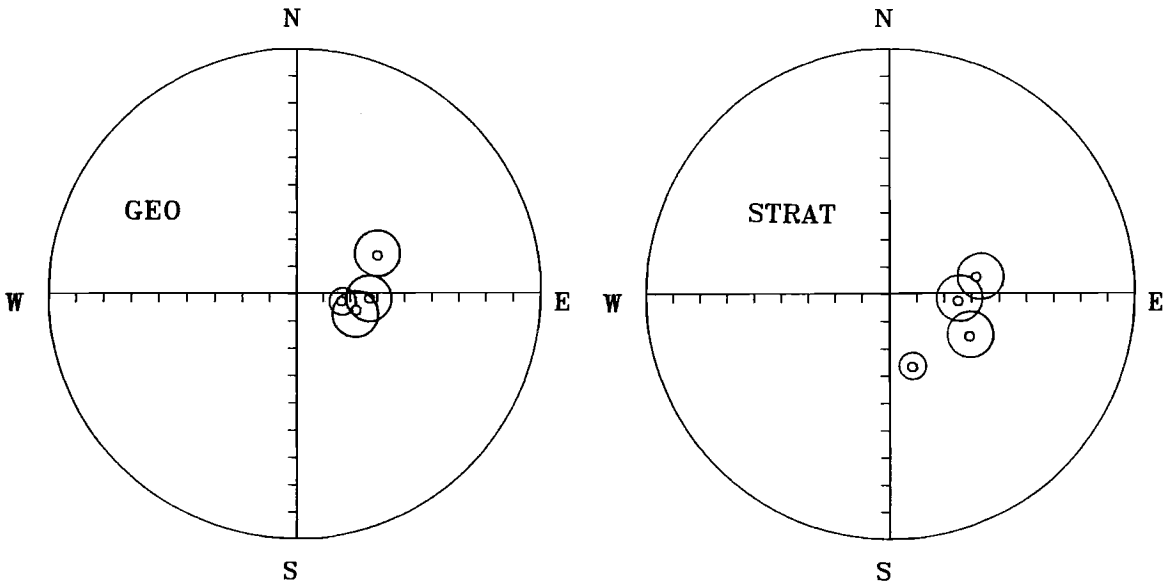


Fig. 14. Stereonets of upper hemisphere showing site mean directions derived from characteristic magnetizations in geographic (Geo) and stratigraphic (Strat) coordinates for Matanuska Formation sites MA1, MA2, MA3, and MA4, north of Castle Mountain Fault. Note dispersion in declination after tilt correction. Shaded regions are Fisher 95% confidence circles.

anisotropies [Graham, 1966] were high (Tables 4 and 5), ranging from 11% for the Chickaloon samples to 18% for the Arkose Ridge samples. Some Arkose Ridge samples had anisotropies as high as 48%.

IRM Acquisition and Thermal Demagnetization of SIRM

Chickaloon and Arkose Ridge samples, that were previously demagnetized in alternating fields up to 100 mT, reached saturation of remanence in DC fields of 50 mT. The Matanuska samples, that were previously demagnetized in alternating fields up to 100 mT, reached saturation in slightly higher fields of 100 mT. Progressive thermal demagnetization of these SIRM's showed a narrower range of unblocking temperatures than the thermal demagnetization of the NRM with unblocking temperatures between 500 °C and 582 °C. The Chickaloon samples were subsequently remagnetized in DC fields of 200 mT and acquired IRM's that were 1000 times stronger than their pre-thermal demagnetization SIRM's. The Matanuska samples, which had shown an intensity increase during routine thermal demagnetization, did not reach saturation of remanence in DC fields as high as 500 mT.

AF Demagnetization of ARM and SIRM

AF demagnetization of ARM's carried by four Chickaloon, ten Arkose Ridge and thirteen Matanuska samples showed distributed coercivity spectra with MDF's near 30 mT. Af demagnetization of saturation IRM's for the same samples generally show a narrower range of coercivities which were lower than those from the ARM's. MDF's for the IRM demagnetizations averaged about 16 mT (Figure 17).

Redeposition-Recompaction Experiment Results

The two disaggregated samples came from two sites (AR6 and AR8). X-ray analyses of these disaggregated samples indicate that the major compositional components are quartz and feldspar with only minor amounts of clay. Examination of these samples under an optical microscope revealed that the samples were completely disaggregated and that separation of the constituent grains only occurred along grain boundaries. Stirring enabled the slurries to acquire an inclination to within 5° of the ambient inclination. In every recompaction experiment, almost all the volume loss occurred at pressures less than 0.05 MPa. At higher pressures, there was only a minimal reduction of volume. Volume loss was between 20% and 60%. This volume decrease behavior is similar to changes in the samples' magnetizations, which showed minor steepening and shallowing of the inclination below 0.05 MPa but little change in inclination above that pressure (Figure 18). After compaction, final magnetizations were within 5° of their original direction. Af demagnetization of the compacted slurries indicate moderate coercivities (MDF = 30 mT) with

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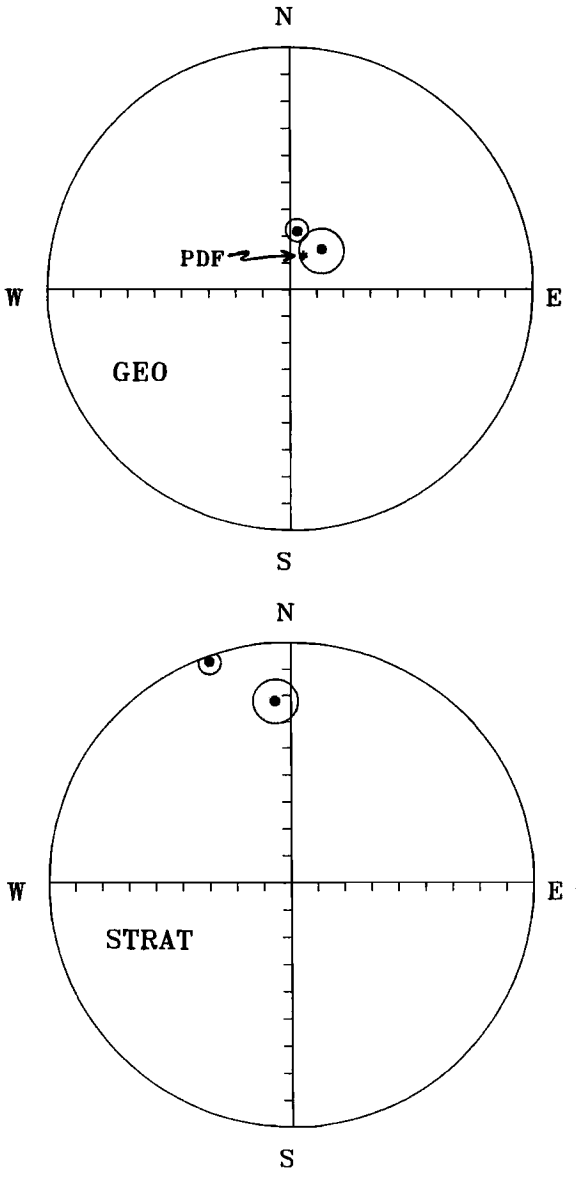


Fig. 15. Stereonets of lower hemisphere showing site mean directions for low unblocking temperature component magnetization in geographic (Geo) and stratigraphic (Strat) coordinates for Matanuska Formation south of Castle Mountain Fault. Shaded regions are Fisher 95% confidence circles. PDF - present day field direction.

unicomponent linear decay of the the magnetic vector into the origin. Af demagnetization after compaction did not change the sample's inclination.

In the redeposition-recompaction experiments, three of the four samples acquired inclinations between 40° and 45°, which was 15° to 20° shallower than the ambient field inclinations during redeposition. The fourth sample acquired a 75° inclination which appeared similar to a steep magnetic field at the mouth of the cryogenic magnetometer's sample access port. The compaction behavior of these redeposited sediments was similar to that of the stirring/recompaction experiments, in that even though the percent volume loss was less (10-15%), most of the volume reduction occurred at pressures below 0.05 MPa. The intensity of the magnetization in the redeposited sediment was almost twice as strong as that in the stirred samples. However, in a fashion similar to the stirred samples, there was minor steepening and shallowing of the inclination below 0.05 MPa but, at higher pressures, the inclination of these samples remained unchanged. Af demagnetization of the three sediments with 40-45° inclinations revealed lower coercivity spectra than that in the stirred samples with two components of magnetization, a lower coercivity component whose inclination was close to the sediments' post-compaction 40° inclination and a very high coercivity (above 50 mT) component that was nearly as steep as the 60° inclination field in which these sediments were deposited. These directions were not only isolated by principal component analysis on linear demagnetization trajectories, but were also suggested by

TABLE 4. Bulk Susceptibility and Percent Anisotropy of Chickaloon Formation, Alaska

Sample	χ_{vol} x 10^{-3} SI	$100(k_{max}-k_{min})/k_{max}$
CH1.6	0.617	8.16
CH2.10	0.466	8.48
CH3.3	1.046	6.00
CH4.9	0.414	18.13
CH5.10	0.461	22.44
CH6.12	0.512	12.10
CH7.6	0.579	7.58
CH8.5	1.202	4.29
CH9.12	0.556	8.54
CH10.12	0.322	11.46
CH12.8	0.766	5.58
CH13.8	0.338	23.34

TABLE 5. Bulk Susceptibility and Percent Anisotropy of Arkose Ridge Formation, Alaska

Sample	χ_{vol} x 10^{-3} SI	$100(k_{max}-k_{min})/k_{max}$
AR1.4	0.403	17.11
AR1.10	0.468	15.45
AR1.11	0.392	17.10
AR4.16	0.519	45.84
AR4.18	0.579	38.48
AR6.6	0.681	21.37
AR8.9	0.371	18.28
AR8.16	0.200	25.90

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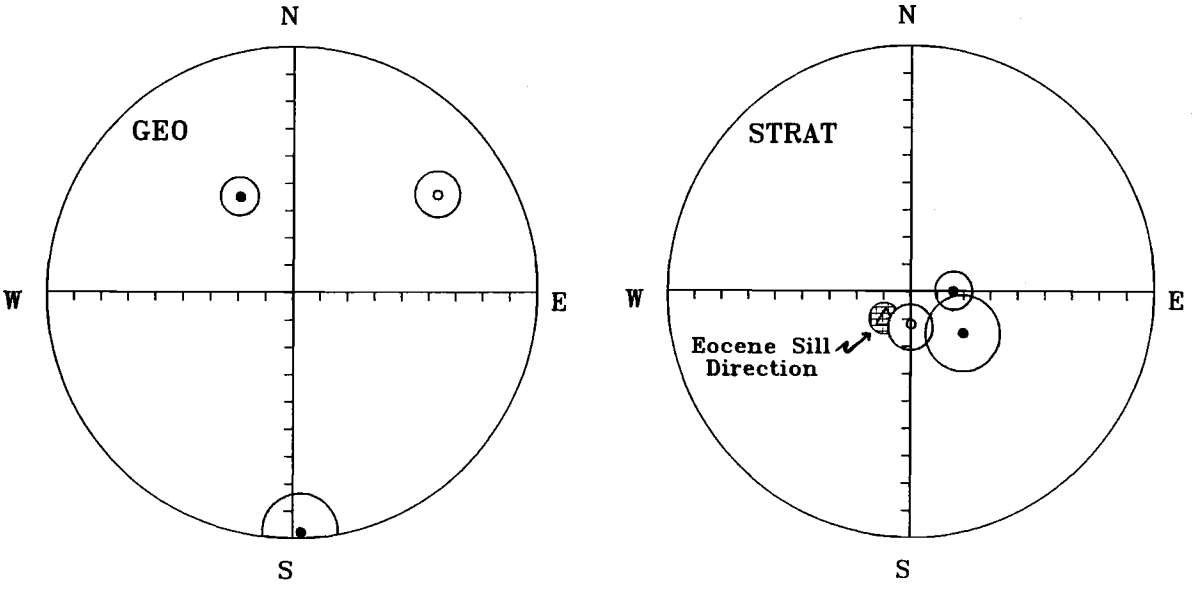


Fig. 16. Stereonets of showing site mean directions in geographic (Geo) and stratigraphic (Strat) coordinates of high unblocking temperature component for Matanuska Formation south of Castle Mountain Fault. Open symbols - lower hemisphere directions; open symbols - upper hemisphere directions; shaded regions - Fisher 95% confidence circles. Eocene sill direction from intrusive rocks in Matanuska Valley [Stamatakos et al., 1988].

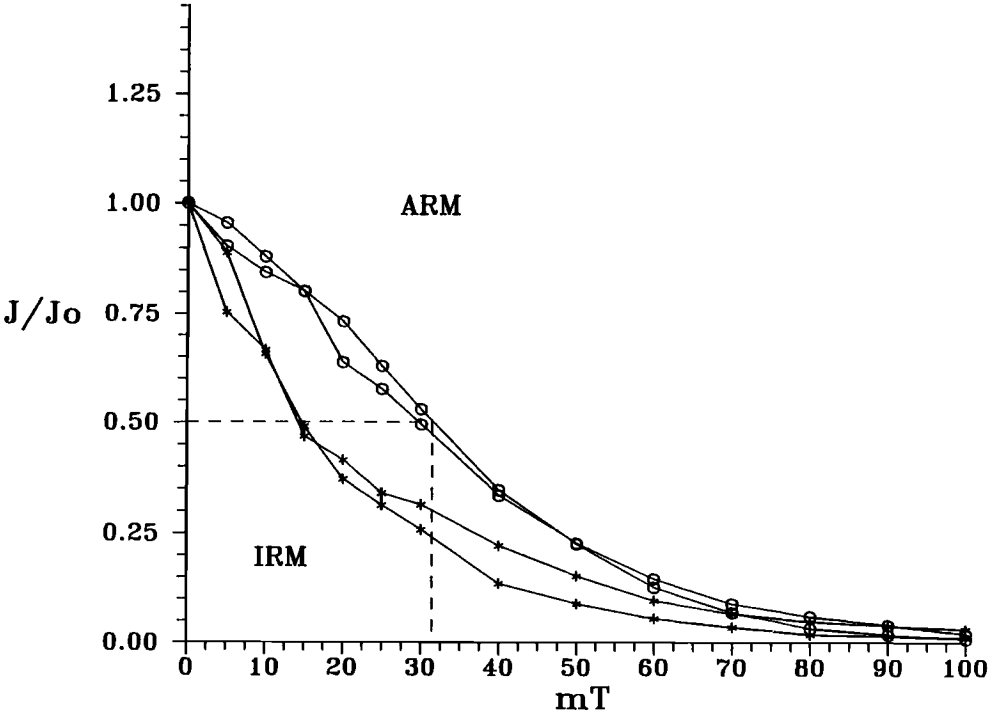


Fig. 17. Normalized intensity diagram showing af demagnetization of ARM and saturation IRM for two Chickaloon Formation samples. Note suggestion of single domain magnetite [Lowrie and Fuller, 1971].

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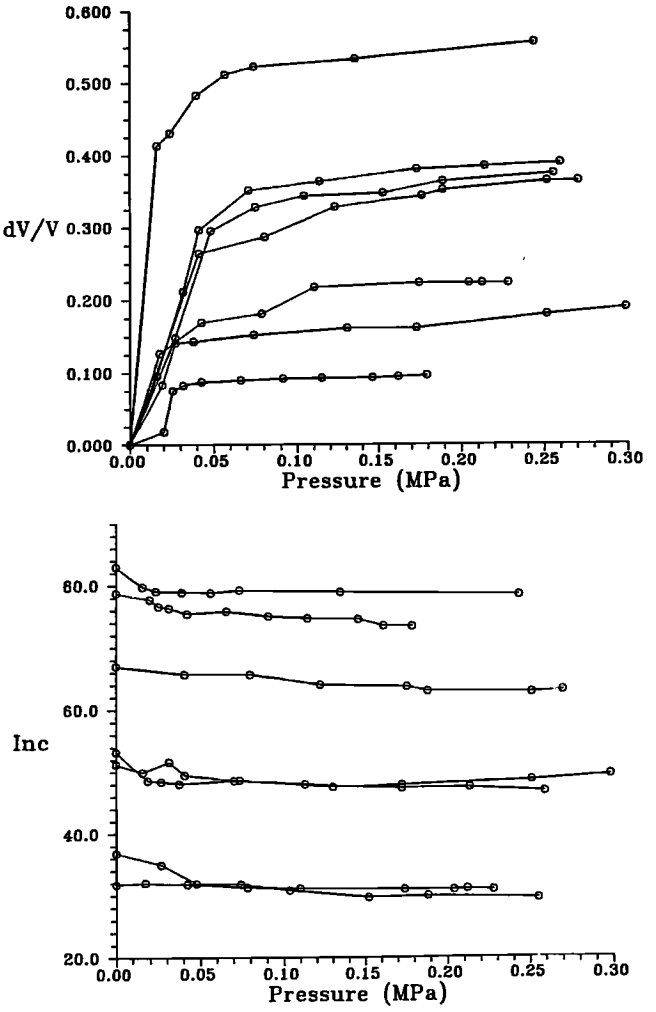


Fig. 18. Results from stirred-slurry/compaction experiments for Arkose Ridge samples. Note that volume (dV/V) and inclination (Inc) changes occur at pressures less than 0.05 MPa. See text for discussion.

remagnetization circles, in which great circle paths fit to the demagnetization directions trend from the 40° inclination direction towards the 60° inclination direction (Figure 19).

Discussion

The IRM acquisition, thermal demagnetization of SIRM, and af demagnetization of both SIRM and ARM indicate that the dominant magnetic phase in all three rock types is single-domain magnetite [Lowrie and Fuller, 1971, Johnson et al., 1975]. The exception is the sample from Arkose Ridge site AR8 which exhibited pseudo-single domain or multidomain behavior. These interpretations are supported by the granulometry experiments which suggest magnetic grains sizes of 1.0 μm or less except for the sample from site AR8 in which 5.0 to 10.0 μm magnetic grains sizes are indicated [Banerjee et al., 1981; King et al., 1983]. Given the silt to fine sand grain size of the sediments, hydrodynamical arguments would suggest larger magnetite grain sizes perhaps indicating that the magnetite may be biogenic [Kirschvink and Chang, 1984] rather than detrital. Intensity increases during thermal demagnetization of the Chickaloon and Matanuska rocks from south of Castle Mountain Fault indicate that magnetite is being produced, perhaps from a reduced form of iron. Because the organic content of these rocks is high, as evidenced by interbedded coals, it is

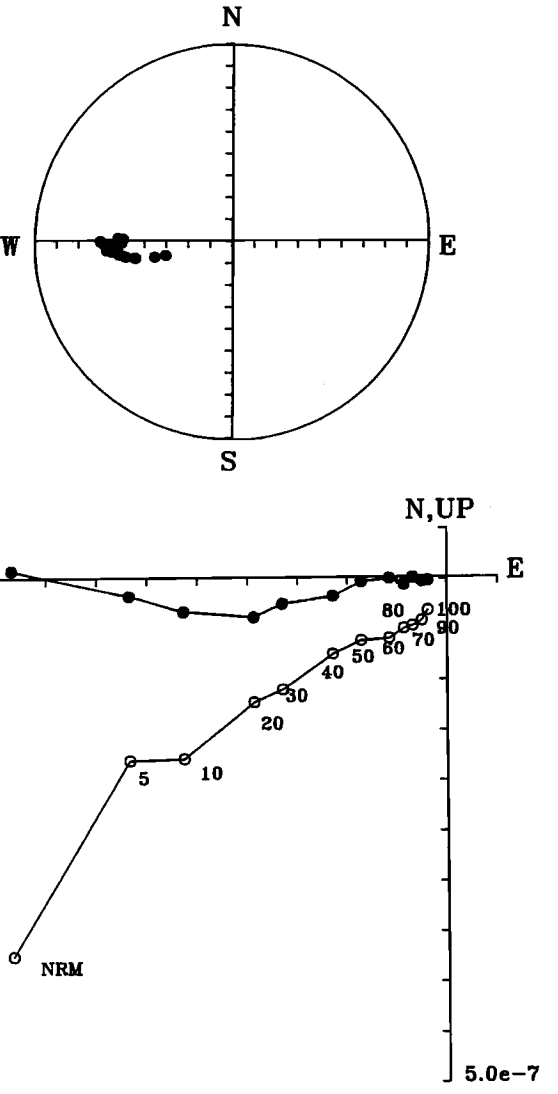


Fig. 19. Orthogonal projection and stereonet plot showing af demagnetization of disaggregated, redeposited, and compacted samples from Arkose Ridge Formation. Note removal of shallow inclinations at higher demagnetization steps. Symbols same as in figure 3. Demagnetization steps in mT.

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possible that this reduced phase is pyrite, pyrrhotite or troilite. These iron sulfides are commonly associated with organic-rich sediments deposited in oxygen deficient environments or as a by-product of hydrothermal alterations [Blanchard, 1939].

The iron sulfide problem does not appear to be as pervasive north of the Castle Mountain fault, where many of the Arkose Ridge and Matanuska samples were thermally demagnetized above 500 °C without any intensity increase. Yet, thermal demagnetization behavior of samples from sites AR5, AR7, and AR9, which were generally the coarsest grained samples, also suggest the presence of pyrrhotite. If the pyrrhotite is, in fact, of hydrothermal origin, then these results imply that hydrothermal alterations were pervasive south of the Castle Mountain Fault but less widespread north of the fault.

Characteristic magnetizations derived from linear demagnetization trajectories for the Chickaloon rocks produced only two site mean directions (CH7 and CH10) which are antipodal when they are corrected for tilt. This suggests that their magnetization is prefolding (pre-Oligocene) in age and assumed to be primary and that secular variation has been time averaged. Because there are results from only two sites (N = 2), the statistical significance of this tilt test cannot be evaluated. However, the similarity of these directions to the directions derived from the demagnetization plane analysis indicate that the demagnetization planes are also successfully isolating a prefolding magnetization. This conclusion is supported by a positive fold test using all seven (N = 7) demagnetization plane-derived site directions [McElhinny 1964] (Figure 9).

Characteristic magnetizations isolated directions for four Arkose Ridge sites (AR3, AR4, AR6, and AR8), two reversed and two normal, which pass a regional tilt test at the 95% confidence level [McFadden and Jones, 1981]. In addition to the fold test, results from site AR2, yield a positive conglomerate test. These results suggest that the magnetization in the Arkose Ridge formation is also pre-folding (pre-Oligocene) in age and assumed to be primary and that secular variation has been time averages.

Because demagnetization plane analysis indicates steepening of the magnetization directions for both the Arkose Ridge and Chickaloon formations during demagnetization, it appears that the characteristic magnetizations have been shallowed 7° to 9° by overprinting and are not accurate records of the earth's field during deposition of the Arkose Ridge and Chickaloon formations.

One problem of demagnetization plane analysis is that if the precision of the secondary magnetization is similar to the precision of the primary magnetization, the intersecting demagnetization planes will yield a direction that is biased towards the secondary magnetization [Schmidt, 1985]. Several observations suggest that this bias does not effect the Chickaloon and Arkose Ridge directions derived from remagnetization planes. When all the Chickaloon or Arkose Ridge demagnetization planes are considered in geographic coordinates, the poles to these planes are scattered and do not fall on a great circle and thus, do not suggest a common post folding secondary magnetization. However, in stratigraphic coordinates, the poles to the remagnetization planes are evenly distributed along a great circle. (Figures 5 and 11). These results suggest that the dispersion of the primary component is much less than that of the secondary component. Moreover, it appears that the demagnetization planes show similar behavior for individual sites (Figure 6) and thus, that individual site directions can be derived from the intersection of demagnetization planes. The validity of this analysis is supported by the similarity between the overall formation direction indicated by the intersection of all the demagnetization planes and the formation mean direction calculated by averaging the demagnetization plane-derived site directions (Tables 1 and 2). In both the Chickaloon and Arkose Ridge data sets, the respective formation directions calculated by either of these two methods are statistically indistinguishable at the 95% confidence level.

Figure 9 reveals that the seven Chickaloon site directions have much higher scatter in declination than in inclination. The folds at these sites do not have steep plunges, and thus, it is not likely that the dispersion in declination is a caused by improperly correcting for plunge and tilt of the beds [Demerest, 1983, MacDonald, 1980]. Equally unlikely is that secular variation has not been time averaged. At most sites a large stratigraphic section was sampled and both magnetic polarities observed. An exception is the fold at site CH2, in which samples were collected from a continuous 1 m thick bed around the core of an anticline. A more probable explanation is that the dispersion in declination is the result of varied amounts of rigid block rotations. A common deformation mechanism in regions dominated by strike-slip faulting is the horizontal rotation of rigid crustal blocks between sets of secondary strike-slip faults antithetic to the enveloping shear [Ron et al., 1984]. Paleomagnetic results from the Eocene sills in the Matanuska Valley indicates 50° of clockwise rotation of the region since the middle Tertiary [Stamatakos et al., 1988], possibly as a result of differential right-lateral shear between the Castle Mountain and Border Ranges faults. Because the Chickaloon rocks are intruded by these sills, they have presumably been rotated clockwise by at least 50°. This is consistent with the Chickaloon data which indicates clockwise rotations of 88°. The spacing of the sites is at the same scale as the

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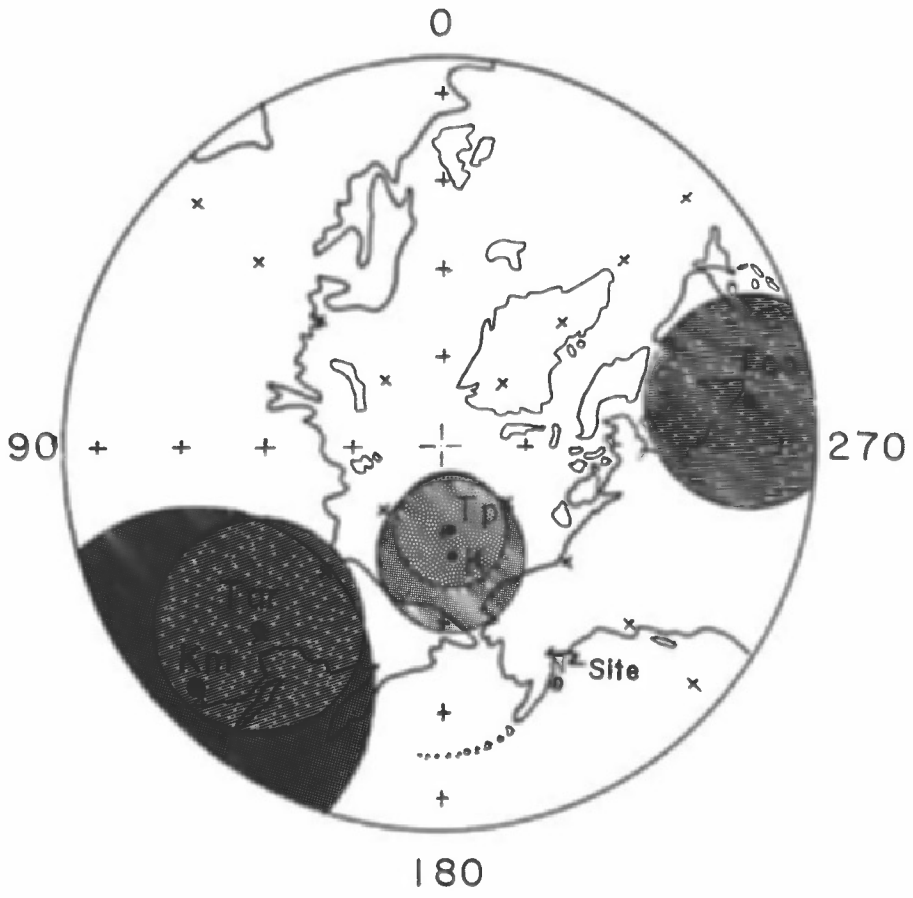


Fig. 20. Polar equal angle projection of paleomagnetic poles for Chickaloon Formation (Tch), Arkose Ridge Formation (Tar), and Matanuska Formation (Km). Tp shows Paleocene pole for North America from Diehl et al. [1983] and K shows Cretaceous pole for North America from Marquis and Globerman [in press]. Shaded regions are 95% confidence limits. Note location in southern Alaska. Limit of projection is latitude 45 °N.

spacing of the rigid blocks [Stamatakos et al., 1988], thus, the resulting pattern of dispersed magnetic declinations is not unexpected.

Because the direction from site CH2 is discordant with the other Chickaloon site directions and because secular variation may not have been time averaged at this site, the demagnetization planes from this site were not used in calculating the paleomagnetic pole for the Chickaloon. Similarly, site AR3 is discordant with the other Arkose Ridge sites and has such an elliptical Bingham confidence oval, possibly because it was located so close to the Castle Mountain fault zone, that its demagnetization planes were discarded from the calculation of the Arkose Ridge paleomagnetic pole. Given these constraints, the best estimate of the paleomagnetic pole of the Chickaloon Formation is 50.5°N, 277.2°E, δ_m = 12.2°, δ_p = 7.7°, and of the Arkose Ridge Formation is 60.4°N, 138.6°E, δ_m = 11.6°, δ_p = 6.4° (Figure 20).

The North American reference pole closest in age to the Chickaloon and Arkose Ridge rocks is the Paleocene pole (81.8°N, 181.4°E , K =20.2) of Diehl et al.[1983]. When the Chickaloon pole is compared to this reference pole, it shows 14.6° ±10.8° of poleward displacement and 88.0° ±20.2° of clockwise rotation. When the Arkose Ridge pole is compared to this reference pole, it shows 12.0° ±10.0 of poleward drift and 47.6° ±20.0° of counterclockwise rotation. These results suggest that there is no paleomagnetically discernible latitudinal offset across Castle Mountain Fault.

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Therefore, this fault system was not a major terrane dispersal system in the Tertiary. These results also suggest that this portion of southern Alaska has migrated poleward 1600 ±1200 km since these sediments were deposited in the Paleocene. This result is in conflict with the previously reported results from the Chickaloon formation [Stone et al., 1982; Stone, 1983] which suggested 5300 ±700 km of poleward displacement.

Results from the Matanuska formation are even more poorly constrained. South of the Castle Mountain Fault the low temperature magnetization component (<350 °C) in geographic coordinates is coincident with the present day field direction (Fig 15) and is interpreted as a viscous (VRM) overprint. The high temperature (>450 °C) component in sites MA7, MA8 and MA10 is prefolding in age and owing to the close proximity of these sampling sites to Eocene sills and their demagnetization behavior, this magnetization appears to be a thermal overprint (PTRM) caused by the Eocene plutonism. This conclusion cannot be tested directly, because a characteristic magnetization in the dike at the baked contact test could not be isolated by line or plane analysis of thermal or af demagnetizations. However, the mean direction of the high temperature magnetic components from sites MA7, MA8, and MA10 (I = 78°, D = 94°) is similar to directions obtained from the Eocene Sills in the Valley (I = -76°, D = 216°) [Stamatakos et al., 1988].

North of the Castle Mountain Fault, the fold test for the Matanuska is inconclusive, although it appears that the directions diverge when the beds are corrected for tilt. However, the observations that the Arkose Ridge rocks pass a regional fold test and that in Boulder Creek there is not an angular contact between the Arkose Ridge and the Matanuska formations [Fuchs, 1980] indicate that if the Matanuska Formation was remagnetized it would have to have been remagnetized before tectonic tilting. Therefore, the direction isolated from the Matanuska should be considered in stratigraphic coordinates. The dispersion, as with the Chickaloon directions, is mainly in declination. Because two of these sites lie just south of the Caribou splay (MA3 and MA4),these sites may have been differentially rotated with respect to one another by strike-slip faulting. Comparison of the Matanuska pole derived from the four sites north of the Castle Mountain Fault with the late Cretaceous North American reference pole (77.7°N, 185°E, A_{95} = 7.7°) tabulated by Marquis and Globerman [in press] yields 25.5° ±18.6° of poleward motion of the Matanuska formation and 49.6° ±32.6° of counterclockwise rotation. This corresponds to a mean poleward displacement of 2880 ±2000 km for the Matanuska since the late Cretaceous.

The counterclockwise rotation of the Arkose Ridge and the Matanuska formations north of the Castle Mountain Fault are consistent with other early Tertiary counterclockwise rotations observed paleomagnetically in southern Alaska [Coe et al., 1985; Hillhouse, 1987] and are presumed to be the result of North American-Eurasian convergence during the late Mesozoic and early Tertiary [Harbert et al., 1987]. Because this counterclockwise rotation may have affected the Chickaloon rocks as well, the total amount of clockwise rotation experienced by these rocks is probably as great as 136°.

An unresolved problem is the nature and origin of the secondary magnetization component observed in the Chickaloon and Arkose Ridge samples. Based on the demagnetization and rock magnetic results this overprint can be interpreted as a chemical (CRM) or a partial thermal remagnetization (PTRM). In geographic coordinates the poles to the demagnetization planes do not fall on a great circle and thus, do not suggest a uniformly directed post-folding overprint. In stratigraphic coordinates, the secondary magnetization shallows the signal. However, if these rocks were remagnetized before they were tilted, and if these rocks were part of a terrane that was continuously drifting poleward, then the remagnetized direction should be as steep as or even steeper than the primary direction. One possible explanation is that the authigenic or hydrothermal pyrrhotite has grown with a preferred orientation due to a preexisting fabric in the rocks caused by either tectonic strain or compaction. Yet, although the percent anisotropies are high, the AMS ellipsoids do not have a regionally systematic orientation suggestive of tectonic strain or primary compaction.

Another possibility is that the secondary magnetization is of opposite polarity to the primary magnetization with an overlapping coercivity or unblocking spectra that inhibit complete isolation of the primary signal. If the two magnetizations are not exactly antipodal, perhaps as the result of secular variation or because the dipole field does not have antipodal polarities [Diehl et al., 1988], then a sample's resultant magnetization will be shallower than either the primary or overprinting directions. Unless the coercivity and unblocking spectra completely overlap, the magnetizations will steepen towards the primary direction during demagnetization. Evidence for these competing magnetizations may be seen in the characteristic directions of the Arkose Ridge Formation (sites AR4, AR6, and AR8), in which the normal polarity directions are steeper than the reversed directions. Moreover, demagnetization plane analysis, which yields steeper formation directions than the characteristic magnetizations, greatly improves the antiparallelism of the reversed and normal polarity directions in both the Chickaloon and Arkose Ridge formations.

The redeposition-recompaction experiments indicate that compaction did not significantly shallow the inclination of these rocks. The

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stronger magnetizations and lower coercivity spectra of the redeposition sediments indicate that the larger, less coercive grains were an important component of the magnetization of these redeposited sediments. Because the redeposited sediments have shallower than expected inclinations, the larger, less coercive grains may have settled with their long axes in the bedding plane during deposition. Only progressive demagnetizations above 50 mT revealed the true primary direction. Unfortunately, the sample chosen for the redeposition experiment was from the only site with pseudo-single and multidomain magnetite. This syn-depositional inclination shallowing model cannot be applied to the very fine grained single domain magnetite in the samples from any of the other sites. However, the stirred-slurry/compaction experiments do suggest that compaction cannot be used to explain the shallower than expected inclinations in these sedimentary rocks.

Tectonic Synthesis

A recent tectonic model by Umhoefer [1987], based on a comprehensive analysis of kinematic indicators for the Pacific basin, implies that the Alaskan superterrane moved 2500 km northward in the Cretaceous as part of the Kula plate, but by 56 Ma this crustal block became attached to North America. Any subsequent northward motion of elements within the superterrane was accommodated by inboard dextral strike-slip faulting. Given these conclusions, the 1600 ±1200 km of northward displacement suggested by the results from the Paleocene Chickaloon and Arkose Ridge Formations may be indicative of the cumulative early Tertiary strike-slip displacement between the stable portion of the North American craton and the Matanuska Valley. The two largest fault systems that could have accommodated this displacement are the Tintina Trench-Northern Rocky Mountain Trench (TT-NRMT) fault zone and the Denali Fault Zone.

Based on systematic mapping of geologic structures, Price and Carmichael [1986] conclude that the TT-NRMT fault system records no more than 450 km of right-hand slip during the Late Cretaceous and early Tertiary. In contrast, Gabrielse [1985] estimated that the amount of offset across the TT-NRMT system in excess of 1000 km from the middle Cretaceous to the Late Eocene or Early Oligocene. This estimate is supported by recent paleomagnetic work on the Upper Cretaceous Carmacks series volcanics in the intermontane Belt in the Whitehorse Trough [Marquis and Globerman, in press] which suggests that these rocks were displaced 1500 ±950 km northward since late Cretaceous time.

Estimates of post collision offset across the Denali fault are equally controversial. Lanphere [1978] indicates that there has been at least 350 km. of offset across this fault system since Paleocene time. However, Csejtey et al. [1982] argue that the Denali fault's 70° curvature along a nonuniform radius inhibits such large displacements. Based on reconnaissance investigations, these authors argue for much smaller offsets, on the order of 10 km or less. More recent work by Nokleberg et al.[1985] suggests that the Maclaren terrane is the western counterpart to the Kluane Schist and the Ruby Range batholith in the southern Yukon and thus, the separation between these elements define at least 400 km of displacement across the Denali fault.

If the maximum estimates for displacements on these faults are correct, then at least some of the observed 14° of latitudinal shift of the Chickaloon and Arkose Ridge rocks can be accounted for by intraterrane strike-slip motion. Critical to this argument is the timing of the movement along these faults and the age of the magnetization in the Arkose Ridge and Chickaloon rocks. Based on the age of the underlying Matanuska Formation [Jones and Grantz, 1961] and the radiometric ages from the upper coal beds [Triplehorn et al., 1984], the Chickaloon is between 70 and 53 Ma. Similarly, the Arkose Ridge Formation rests on metamorphic rocks that are 60 Ma and is capped by volcanics that are roughly 50 Ma. Therefore, if the Denali, TT-NRMT, and other smaller fault systems are the mechanism by which the Chickaloon and Arkose Ridge rocks were translated northward in the Paleocene, then this motion must have occurred between the deposition of these sediments, after 60 Ma, and the extrusion of the 50 Ma volcanics in the northern Talkeetna Mountains, which have paleomagnetic poles nearly coincident with Eocene reference poles for North America [Hillhouse et al., 1985].

In their paleomagnetic study of these volcanics, Hillhouse et al. [1985] tested whether large displacements on the TT-NRMT fault system could account for the observed counterclockwise rotations in southern Alaska, by fitting a small circle with an apical half-angle of 22° to the Tintina fault, calculating the rotation about this pole needed to accommodate 1000 km of dextral offset, and then rotating their mean Alaskan site back to its pre-faulted position. Their results predicted 37° of counterclockwise rotation of declination which is close to the 40-50° of observed declination rotation, but, they rejected this model because it also predicted a paleolatitudinal anomaly of 5° which does not agree with their paleomagnetic result from the Talkeetna Mountains. In a similar calculation, using the maximum of 1500 km of cumulative offset for the Denali and the TT-NRMT, and assuming that the two faults have similar small circle trajectories, rotation of the Matanuska Valley around the small-circle pole of Price and Carmichael [1986], predicts a latitudinal offset of only 7°. These calculations suggest that unless these fault traces trended more northerly in the Paleocene, and thus were better aligned to

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produce more paleomagnetically detectable displacements, or that the structural interpretations have underestimated the amount of slip recorded by these faults, the actual amount of latitudinal offset for the Arkose Ridge and Chickaloon formations, based on our upper 95% confidence limit, is closer to the minimum estimate of 3°. Reconstruction of the North America in the late Cretaceous and early Tertiary, based on the apparent pole wander path (APWP) of Irving and Irving [1982], indicates that although the trend of the northern Cordillera continental margin was almost due north in the late Cretaceous, by the Paleocene it was northwesterly, similar to its trend today. Therefore, some of the 12.5° of additional latitudinal offset seen in the Matanuska sites north of the Castle Mountain Fault, when compared to the Arkose Ridge data, could have been caused by strike-slip faulting parallel the continental margin between the late Cretaceous and Paleocene.

Conclusion

Paleomagnetic and rock magnetic results from the Paleocene Arkose Ridge Formation north of the Castle Mountain Fault and the Chickaloon Formation south of this fault indicate that these sedimentary rocks have a prefolding magnetization carried by fine-grained single-domain magnetite. These magnetizations indicate that there is no paleomagnetically discernible offset across the Castle Mountain Fault and suggests that this fault system did not act as a major terrane dispersal mechanism in the early Tertiary. In addition, both the Chickaloon and Arkose Ridge paleomagnetic poles indicate approximately 14° of poleward motion with respect to North America. Laboratory compaction experiments suggest that post-depositional compaction cannot be used explain the shallow Arkose Ridge signal. Thus, the 1600 ±1200 km of displacement predicted by these data may be indicative of the the latitudinal component of the cumulative dextral motion between 60 my and 50 my on strike-slip faults subparallel to the continental margin. However, based on the orientation and maximum structurally discernible displacements of two of these faults, the Tintina and Denali fault systems, only half of the paleomagnetically observed displacement can be explained.

In addition to the inclination data, the paleomagnetic results from these Paleocene Formations indicate that the Arkose Ridge rocks were rotated counterclockwise 48° and that the Chickaloon rocks were rotated clockwise at least 88° and as much as 136° since the Paleocene. The counterclockwise rotation of the Arkose Ridge rocks is consistent with other paleomagnetically observed rotations in southern Alaska. The clockwise rotation of the Chickaloon rocks is consistent with the paleomagnetically observed 50° of clockwise rotations of the Eocene sills in the eastern portion of the Matanuska Valley [Stamatakos et al., 1988]. These clockwise rotations can be explained as the result of rigid block rotations located within a broad shear zone created by right-lateral strike-slip faulting along the Castle Mountain and Border Ranges fault systems.

Paleomagnetic and rock magnetic analysis of the Cretaceous Matanuska Formation indicate that south of the Castle Mountain Fault these rocks contain only secondary viscous and thermal overprints. North of the fault, the Matanuska rocks carry a magnetization that, on the basis of the Arkose Ridge fold test, may be prefolding in age. This magnetization indicates that these rocks originated 2800 ±2000 km south of their present latitudinal position with respect to North America. Based on similar arguments to the one outlined for the Arkose Ridge and Chickaloon formations, these results suggest that either there was rapid motion of the Peninsular terrane in the Late Cretaceous, possibly along more northerly trending strike-slip faults which were better aligned to yield paleomagnetically detectable offset, or that like the Chickaloon and Arkose Ridge rocks, the actual amount of latitudinal displacement of the Matanuska formation is closer to its lower 95% confidence limit of 800 km.

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References

Anson, G. L., and Kodama, K.P., Compaction-induced inclination shallowing of the post-depostional remanent magnetization in a synthetic sediment, Geophys. J. R. astr. Soc., 88, 673-692, 1987.

Banerjee, S.K., King, J., and Marvin, J., A rapid method for magnetic granulometry with applications to environmental studies, Geophysical Research Letters, v. 8, p. 333-336, 1981.

Barnes, F.F., Geologic map of the lower Matanuska Valley, Alaska, scale 1:63,360, U.S. Geological Survey Miscellaneous Geological Investigations Serial Map I-359., 1962.

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Barnes, F.F., and Payne, T.G., Wishbone Hill district, Matanuska coal field, Alaska, U.S. Geological Survey Bulletin 1016, 88 p., 1956.

Blanchard, R., Paragenesis of pyrrhotite, Economic Geology, v. 33, p. 218, 1939.

Bruhn, R.L., and Pavlis, T.L., Late Cenozoic deformation in the Matanuska Valley, Alaska; three-dimensional strain in a forearc region, Geological Society of America Bulletin, v. 92, Part 1, p. 282-293, 1981.

Capps, S.R., Martin, G.C., Mertie, J.B., Jr., Katz, F.J., and Landes, K.K., Geologic map of upper Matanuska Valley, Alaska, U.S. Geological Survey Bulletin 791, plate 2, scale 1:62,500, 1927.

Clardy, B.I., Origin of the lower and middle Tertiary Wishbone and Tsadaka formations, Matanuska Valley, Alaska [M.S.thesis]: Fairbanks, University of Alaska, 50 pp., 1974.

Coe, R.S., Globerman, B.R., Plumley, P.W., and Thrupp, G.A., Paleomagnetic results from Alaska and their tectonic implications: in David G. Howell, ed., Tectonostratigraphic terranes, Pacific northeast quadrant: Houston, Texas, Circum-Pacific council for energy and mineral resources, p. 85-108, 1985.

Coney, P.J., Mesozoic-Cenozoic Cordillera plate tectonics, Geological Society of America, Memoir 152, P. 33-50, 1978.

Coney, P.J., Jones, D.L., and Monger, J.W.H., Cordilleran suspect terranes, Nature, v. 288, p. 33-50., 1980

Csejtey, B., Jr., Tectonic implications of a late Paleozoic volcanic arc in the Talkeetna Mountains, South-Central Alaska, Geology, v.4, p.49-52, 1976.

Csejtey, B., Jr., Cox., D.P., Evarts, R.C., Striker, G.D., and Foster, H.L., The Cenozoic Denali fault system and the Cretaceous accretionary development of southern Alaska, Journal of Geophysical Research, v. 87, p. 3741-3754, 1982.

Csejtey, B., Jr., Nelson, W.H., Jones, D.L., Silberling, N.J., Dean, R.M., Morris, M.S., Lanphere, M.A., Smith, J.G., and Silberman, M.L., Reconnaissance Geologic Map and Geochronology, Talkeetna Mountains quadrangle, northern part of Anchorage quadrangle, and southwest corner of Healy quadrangle, Alaska, scale 1:250,000, U.S. Geological Survey Open File Report, 78-558-A, p. 1-62, 1978.

Davis, G.A., Monger, J.W.H., and Burchfiel, B.C., Mesozoic construction of the Cordilleran 'collage', central British Columbia to central California, in Mesozoic Paleogeography of the Western United States, Pacific Coast Paleogeology Symposium 2, edited by D. G. Howell and K. A. McDougall, Society of Economic Paleontologists and Mineralogists, p. 1-32, 1978.

Demarest, H.H., Jr., Error analysis for the determination of tectonic rotation from paleomagnetic data, Journal of Geophysical Research, v. 88, p. 4321-4328, 1983.

Detterman, R.L., Plafker, G., Hudson, T., Tysdal, R.G., and Pavoni, N., Geology and surface features along the Talkeetna segment of the Castle Mountain-Caribou Fault system, Alaska, U.S. Geological Survey Miscellaneous Field Studies Map MF-734; scale 1:63,360, 1974.

Diehl, J.F., Beck, M.E., Jr., Beske-Diehl, S., Jacobson, D., and Hearn, B.C., Jr., Paleomagnetism of the Late Cretaceous-early Tertiary north-central Montana alkalic province, Journal of Geophysical Research, v. 88, p. 10,593-10,609, 1983.

Diehl, J.F., McClannahan, K.M., and Bornhorst, T.J., Paleomagnetic Results from the Mogollon-Datil volcanic field, southwestern New Mexico, and a refined mid-Tertiary reference pole for North America, Journal of Geophysical Research, v. 93, p. 4869-4879, 1988.

Fisher, R.A., Dispersion on a sphere, Royal Society of London Proceedings, ser. A, v. 217, p. 295-305, 1953.

Fuchs, W.A., Tertiary tectonics history of the Castle Mountain-Caribou fault system in the Talkeetna Mountains, Alaska, Ph.D thesis, Salt Lake City, Utah, University of Utah., 1980.

Gabrielse, H., Major dextral transcurrent displacements along the northern Rocky Mountain Trench and related lineaments in north-central British Columbia, Geological Society of America Bulletin, v. 96, p. 1-14, 1985.

Graham, J.W., The stability and significance of magnetism in sedimentary rocks, Journal of Geophysical Research, v. 58, p. 243-260., 1949.

Graham, J.W., Significance of magnetic anisotropy in Appalachian sedimentary rocks, in J.S. Steinhart and T.J. Smith (eds) The Earth Beneath the Continents, Geophysical Monograph 10; National Academy of Sciences-National Research Council Publication 1476, 627-648, 1966.

Grantz, A., Geologic map of Talkeetna Mountains (A-1) quadrangle and south third of Talkeetna Mou
ntains (B-1) quadrangle, Alaska, U.S. Geological Survey Miscellaneous Geological Investigations Map I-343, scale 1:48,000, 1960.

Grantz, A., and Jones, D.L., Stratigraphy and age of the Matanuska formation, south-central Alaska, Geological Survey Research, *in* Short Papers in the Geological Sciences, U.S. Geological Survey Professional Paper 400-B, p B347-B350., 1960.

Grantz, A., and Wolfe, J.A., Age of the Arkose Ridge Formation south-central Alaska, American Association of Petroleum Geologists Bulletin, v. 45, p. 1762-1765, 1961.

Gromme, S., and Hillhouse, J.W., Paleomagnetic evidence for northward movement of the Chugach Terrane, southern and south-western Alaska, U.S. Geological Survey, Circular 823-B, p. B70-B72., 1981.

Halls, H.C., A least-square method to find remanence directions from converging remagnetization circles, Geophysical Journal of

10.1029/GM050p0151. Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GM050p0151 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the Royal Astronomical Society, v. 45, p. 297-304, 1976.

Harbert, W., Frei, L.S., Cox, A., and Engebretson, D.S., Relative motions between Eurasia and North America in the Bering Sea Region, Tectonophysics, v. 134, p. 239-261, 1987.

Hillhouse, J.W, Accretion of southern Alaska, Tectonophysics, v. 139, p. 107-122, 1987.

Hillhouse, J.W., Gromme, C.S., Limits to the northward drift of the Paleocene Cantwell Formation, central Alaska, Geology, v. 10, p. 552-556, 1982.

Hillhouse, J.W., Gromme, C.S., and Csejtey, B., Jr., Tectonic implications of paleomagnetic poles from lower Tertiary volcanic rocks, south-central Alaska, Journal of Geophysical Research, v. 90, p. 12,523-12,535, 1985.

Imlay, R.W., New genera of early Cretaceous (Albian) ammonites from Alaska, Journal of Paleontology, v. 33, p. 179-185, 1959.

Irving E., and Irving G.A., Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana, Geophysical Surveys 5, p. 141-188., 1982.

Johnson, H.P., Lowrie, W., Kent, D.V., Stability of anhysteretic remanent magnetizations in fine and coarse magnetite and maghemite particles, Geophysical Journal of the Royal Astronomical Society, v. 41, p. 1-10, 1975.

Jones, D.L., and Grantz, A., Upper Cretaceous (Campanian and Maestrichtian) ammonites from southern Alaska, Geological Survey Professional Paper 432, 49 p., 1963.

Jones, D.L., and Grantz, A., Cretaceous ammonites from the lower part of the Matanuska Formation, southern Alaska, U.S. Geological Survey Professional Paper 547, 49 p., 1967

Jones, D.L., Irwin, P., and Oveushine, T., SE Alaska - a displaced continental fragment?, U.S., Geological Survey Professional Paper, 800-B, B211-217., 1972.

King, J., Banerjee, S.K., Marvin, J., and Ozdemir, O., A comparison of different magnetic methods for determining the relative grain size of magnetite in natural materials: Some results from lake sediments, Earth and Planetary Science Letters, v. 59, p. 404-419, 1983.

King, P. B., Tectonic Map of North America, U.S. Geological Survey map, scale 1:5,000,000, 1969.

Kirschner, C.E., and Lyon, C.A., Stratigraphic and tectonic development of Cook Inlet Basin Petroleum Province, in Max G. Pitcher, ed., Arctic Geology: Tulsa, Oklahoma, American Association of Petroleum Geologists, Memoir 19, p. 396-407, 1973.

Kirschvink, J.L., The least-square line and plane and the analysis of paleomagnetic data, Royal Astronomical Society Geophysical Journal, v. 62, p. 699-718, 1980.

Kirschvink, J.L. and Chang, S-B R., Ultrafine-grained magnetite in deep-sea sediments: Possible bacterial magnetofossils, Geology, 12, 559-562, 1984.

Lahr, J.C., Page, R.A., Stephens, C.D., and Fogleman, K.A., Sutton, Alaska, earthquake of 1984: Evidence for activity on the Talkeetna segment of the Castle Mountain fault system, Seismological Society of America Bulletin, v. 76, p. 967-982, 1986.

Lanphere M.A., Displacement history of the Denali fault system, Alaska and Canada, Canadian Journal of Earth Sciences, v. 15, p 817-822, 1978.

Lowrie, W. and Fuller, M., On alternating field demagnetization characteristics of multidomain thermoremanent magnetizations in magnetite, Journal of Geophysical Research, v. 76, 6339-6349, 1971.

MacDonald, W.D., Net tectonic rotation, apparent tectonic rotation and the structural tilt correction in paleomagnetic studies, Journal of Geophysical Research, 85, 3659-3669, 1980

Martin, G.G., and Katz, F.V., Geology and coal fields of the lower Matanuska Valley, Alaska, U.S. Geological Survey Bulletin 500, 88 pp., 1912.

McElhinny, M.W., Statistical Significance of the fold test in paleomagnetism, Geophysical Journal of the Royal Astronomical Society, v. 8, p. 338-340, 1964.

McFadden P.L., and Jones, D.L., The fold test in paleomagnetism, Royal Astronomical Society Geophysical Journal, v. 67, p. 53-58, 1981.

Monger, J.W.H., and Price, R.A., Geodynamic evolution of the Canadian Cordillera-progress and problems, Canadian Journal of Earth Sciences, v. 16, p. 770-791., 1979.

Moore, J.C., Byrne, T., Plumley, P.W., Reid, M., Gibbons, H., and Coe, R.S., Paleogene evolution of the Kodiak Island, Alaska, consequences of ridge-trench interaction in a more southerly latitude, Tectonics, v. 2, p. 265-293, 1983.

Nokleberg, W.J., Jones, D.L., and Silberling, N.J., Origin and tectonic evolution of the Maclaren and Wrangellia terranes, eastern Alaska Range, Alaska, Geological Society of America Bulletin, v. 96, p 1251-1270, 1985.

Onstott, T.C., Application of the Bingham distribution function in paleomagnetic studies, Journal of Geophysical Research, v. 85, p. 1500-1510, 1980.

Panuska, B.C., Paleomagnetic evidence for a post-Cretaceous accretion of Wrangellia, Geology, v. 13, p. 880-883., 1985.

Payne, T.G., Mesozoic and Cenozoic tectonic elements of Alaska, U.S. Geological Survey Miscellaneous Geological Investigations, Map I-84, 1955.

Plumley, P.W., A paleomagnetic study of the Prince William terrane and Nixon fork terrane, Alaska, PhD Dissertation, University of California, Santa Cruz, 190 p., 1984

Plumley, P.W., Coe, R.S., Byrne, T., Paleomagnetism of the Paleocene Ghost Rocks

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Formation. Prince William Terrane, Alaska, Tectonics, v. 2, P 295-314, 1983.

Price, R.A., and Carmichael, D.M., Geometric test for late Cretaceous-Paleogene intrcontinental transform faulting in the Canadian Cordillera, Geology, v. 14, p. 468-471, 1986.

Richter, D.H., and Jones, D.L., Structure and stratigraphy of eastern Alaska Range, Alaska, in Arctic Geology, American Association of Petroleum Geology, Memoir 19, p 408-420, 1973.

Ron, H., Freund, R., Garfunkel Z., and Nur, A., Block rotation by strike-slip faulting: Structural and paleomagnetic evidence, Journal of Geophysical Research, v.89, p.6256-6270,1984.

Silberman, M.L., and Grantz A., Paleogene volcanic rocks of the Matanuska Valley area and displacement history of the Castle Mountain fault in Coonrad, W.L., and Elliot, R.I., eds., The U.S. Geological Survey in Alaska: accomplishments during 1981, U.S. Geological Survey Circular 0868, p. 82-86, 1984.

Stone, D.B., Concordance and conflicts in Alaska paleomagnetic data, in D.G. Howell, et al, eds., Proceedings of the Circum-Pacific Terrane conference, Stanford University Publications, Geological Sciences, v. 18, p. 186-189, 1983.

Stone, D.B., and Packer, D.R., Tectonic implications of Alaska Peninsula paleomagnetic data, Tectonophysics, v. 37, p. 183-201, 1977.

Stone, D.B., and Packer, D.R., Paleomagnetic data from the Alaskan Peninsula, Geological Society of American Bulletin, v 90, p 545-560, 1979.

Stone, D.B., Panuska, B.C., and Packer, D.R., Paleolatitude versus time for southern Alaska, Journal of Geophysical Research, v. 87, p. 3697-3707, 1982.

St. Amand, P., 1957, Geological and geophysical synthesis of the tectonics of portions of British Columbia, the Yukon Territory, and Alaska, Geological Society of America Bulletin, v. 68, p. 1348-1370, 1957.

Stamatakos, J.A., Kodama, K.P., and Pavlis, T.L., Paleomagnetism of Eocene plutonic rocks, Matanuska Valley, Alaska, Geology, v. 16, p 618-622, 1988.

Thrupp, G.A., and Coe, R.S., Early Tertiary paleomagnetic evidence and the displacement of southern Alaska, Geology, v.14, p. 213-217, 1986.

Triplehorn D.M., Turner, D.L., and Naeser, C.W., Radiometric age of the Chickaloon formation of south-central Alaska: Location of the Paleocene-Eocene boundary, Geological Society of America Bulletin, v. 95, p. 740-742, 1984.

Tucker, P., Stirred remanent magnetizations: a laboratory analogue of post-depositional realignment, Journal of Geophysical Research, v. 48, 153-157, 1980.

Umhoefer, P.J., Northward Translation of "Baja British Columbia" along the Late Cretaceous to Paleocene Margin of western North America, Tectonics, v. 6, p. 377-394, 1987.

Waring, G.A., Geology of the Anthracite Ridge coal district, Alaska, U.S. Geological Survey Bulletin, v. 867, 57 p., 1936.

Wolfe, J.A., Hopkins, D.M., and Estella, B.L., Tertiary stratigraphy and paleobotany of Cook Inlet region, Alaska, U.S. Geological Survey professional paper 398-A, p. A1-A29, 1966.

Zijderveld, J.D.A., AC demagnetization of rocks: Analysis of results, in Methods of Paleomagnetism, edited by D.W. Collinson, K.M. Creer, and S.K. Runcorn, pp. 254-286, Elsevier, Amsterdam, 1967.

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